

MAGNETOM

Troubleshooting Guide System

RF System

Harmony syngo MR
Symphony syngo MR
Sonata syngo MR
Trio
Upgrades to syngo MR
from Impact/Expert
from Vision

Document Version

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1.1 General

In this part you will find instructions for the RF-System, containing the strategy and test procedures

One of the essential prerequisites for excellent image quality of a MR system is a correct and stable RF system.

However, due to the complexity of a MR system, image quality is influenced by a great variety of components and functions, such as

- Communication systems (e.g. CAN bus, general data transmission,...)
- Control systems (e.g. Host, MMC,...)
- Image generating systems (e.g. RF system, gradient system,...)
- Image acquisition, processing and archiving systems (e.g. MMC, MRIR,...)
- Software performance

The main tool for RF troubleshooting are the **RF tests**, available in the Service Software platform "**Test Tools**":

The RF test includes communication tests, measurements of the stability, linearity and signal levels of RF components using loop tests, which step by step include more components of the TX/RX path.

In case one of the tests fail, a diagnosis will show you the FRU's responsible.

Additional hardware tests are described in the section on (→ Procedures / Page 34).

In the manual "Replacement of Parts" the necessary procedures, adjustments and final checks when replacing a FRU are described.

The following is an attempt to provide the CSE with a shortcut that starts with the symptom and ends with identifying the respective FRU in the RF system by using the corresponding test.

1.2 Proposed general order of RF Test Procedures

Step	Service action	References
0.	Prerequisites/Preliminary tests	
0.1.	Run Test Tool - Communication Test (MMC/MPCU-, MMC/PCI/CAN-, RF-Comm/Status)	Test Tools / Online Help for specific test
0.2.	Run Tests Tools - MMC tests (MC4C40, RX4)	
0.3.	Run Test Tools - Imager tests	
1.	RF tests (a full test takes about 90 minutes)	
1.1.	Run Test Tools RF tests	Test Tools / Online Help for specific test
1.2.	Perform specific procedures in case of doubt	RF procedures
2.	Perform Tune-up/QA measurements	
To verify the success of the repair/replacement of RF components, specific adjustment must be performed (refer to Replacement of Parts document). Finally, a complete run of QA is recommended.		Tune-up / Online Help QA / Online Help



Please ensure that the system is powered on at least for 20 minutes before performing any measurements and/or procedures using the Service Software procedures (TestTools/RF Tests).

1.3 Proposed sequence of Test Procedures - FRU-based

The following tests are recommended to verify the function of a specific Field Replaceable Unit (e.g. in case of problems after FRU replacement).

1.3.1 RFSU

RFSU	
FRU	Test Procedures
TX-Module	Run Test Tools RF / MOD/REC
RX-Module	Run Test Tools - RF / MOD/REC - RF / LCCS

1.3.2 RFPA

1.3.2.1 RFPA (Ha/ Sy/ Sonata)

RFPA (Ha/ Sy/ Sonata)	
FRU	Test Procedures
RFPA (DORA/CORA)	Run Test Tools - RF / RFPA / Comm/Status - RF/ RFPA1/2

1.3.2.2 RFPA Trio

RFPA Trio	
FRU	Test Procedures
RFPA 8kW	LED "Overload" must be OFF. If ON after measurement, run Test Tools: - RF / RFPA2 - RF / Cable

RFPA Trio	
FRU	Test Procedures
RFPA 35kW (Tube Amp) PS Deck	Run Test Tools - RF / RFPA / Comm/Status - RF/ RFPA1
RF Deck	
PA tube IPA tube	
CAN Bus Controller	Run Test Tools RF / RFPA / Comm/Status

1.3.3 RFAS

RFAS	
FRU	Test Procedures
LCCS D11	Run Test Tools RF / LCCS
BCCS D5	Run Test Tools RF / RFAS
TALES	
TAS	Run Test Tools - RF / RFAS - RF / Cable
SAMI	Run Test Tools RF / Cable
BTB	Run Test Tools - RF / BTB - RF / BC

1.3.4 RFCI

RFCI	
FRU	Test Procedures
Power Supply	Run Test Tools - RF / BTB - RF / BC - RF / RFCI
Motherboard D14, 57 73 085	Run Test Tools RF / RFCI
DYSCON	Run Test Tools RF / RFCI PIN Diodes
CAN Module	Run Test Tools RF / RFCI

1.3.5 PIN Booster (Trio)

PIN Booster (Trio)	
FRU	Test Procedures
PIN Booster	Run Test Tools RF / RFCI PIN Diodes (Local/Body)

1.3.6 Coils

Coils	
FRU	Test Procedures
Body Coil Ha/ Sy/ Sonata water cooled) (Refer to (→ Body Coil (Ha/ Sy/ Sonata) / MR-000.841.14))	For NUMARIS 3.5: - Refer to Update Instruction MR 009/99/1, Body Coil decoupling check. For NUMARIS 4 >VA12A: - Run TestTools / RF / BTB and BC tests interactive. » The spec for the transmission value is: Transmission < -14 dB; after proper adjustment of BC it should be lower than -20 dB. only in case of problems with BC overtemperature: - Run Supervision D9 Test (→ Supervision D9/Grad.coil Test Procedure / MR-000.840.16).
Body Coil Type 2 (Refer to (→ Body Coil Type-2 / MR-000.841.14))	Run Test Tools RF / BTB- and BC
Body Coil Trio (BCT) (Refer to (→ Body Coil Trio (BCT) / MR-000.841.14))	
Local coils	in case local coil replacement did not solve the problem: - RF / RFCI PIN Diodes - RF / LCCS plug_nn (Service Plug) interactive

1.4 RF-related Hardware Errors, Communication Errors

Symptoms	Service action	References
Specific error message Breakdown of a specific component (FRU) Adjustments/measurements fail	■ Check error messages in the Event Log window (from the Service Software homepage) for remedy. ■ Check error LED's. ■ Check power supply units, fuses. ■ Consider the possibility of monitoring errors -> check error transmission lines (FOC's, optical transmitter and receiver). ■ Replace defective component; or replace components of equal type (if used in the system) with each other. ■ Perform specific RF component tests.	(→ Interactive Functional Description / MR-000.850.05) of the corresponding FRU's Wiring Diagrams / RF-system RF procedures
Communication error	Run Test Tools: ■ RF / Comm/Status ■ MMC / PCI/CAN, Comm/Status Check communication lines (CAN-units of RF system are): ■ RFSU / TX-module ■ RFPA ■ RFAS / BTB,LCCS, RFCI	Test Tools / Online Help
Noise image only	■ Check cables/connections in the RF path ■ Check signal level of RF components with scope	(→ Procedures / Page 34)

1.5 RF-related Image Quality problems

Area of application:

- Specific QA procedures such as S/N DIP, Calc Artefacts, RF Verify or Stability Check fail and / or
- General Image Quality problems

Symptoms	Service action	References
Ghosting in phase encoding direction independent of orientation	■ Run the RF tests » Check the results of stability evaluation	Test Tools / RF, Online Help
Low S/N	■ Run the RF tests » Check the results of stability and linearity evaluation ■ Eliminate possible spike sources	
Slice profile broadening	■ Run the RF tests » Check the results of linearity evaluation ■ Check modulator SSB1 signal with scope	Tune-up, On-line Help; (→ Procedures / Page 34)
IQ problem depends on specific coil	■ Check the local coil visually for damages ■ Run the RF tests for the specific RF-path ■ Check the tuning/detuning signals with the help of the Service Plug (RFCI/LCCS interactive tests) ■ Use another coil of the same type	(→ Procedures / Page 34)
IQ problem depends not on specific coil	Run the RF tests especially: ■ MOD/REC1,2 ■ RFPA ■ RFAS	Test Tools / RF, Online Help

1.6 Service strategy RFPA 35 KW (Tube Amp)

1.6.1 General Safety Information for RFPA 35 kW



WARNING

High voltage!

Hazard of electric shock or burn !

Even when the MR system is switched off, mains voltage at the mains input of the amplifier is present.

- » Before you work on the amplifier, switch-off the circuit breakers CB2 and then CB1 at the amplifier and at least circuit breaker F103 in the mains box (ACC) and ensure against accidental switch-on.
- » Observe the warning information on the amplifier.
- » After the amplifier has been turned off, allow at least ten minutes for the high voltage filter capacitors to discharge fully before removing any of the cover plates.

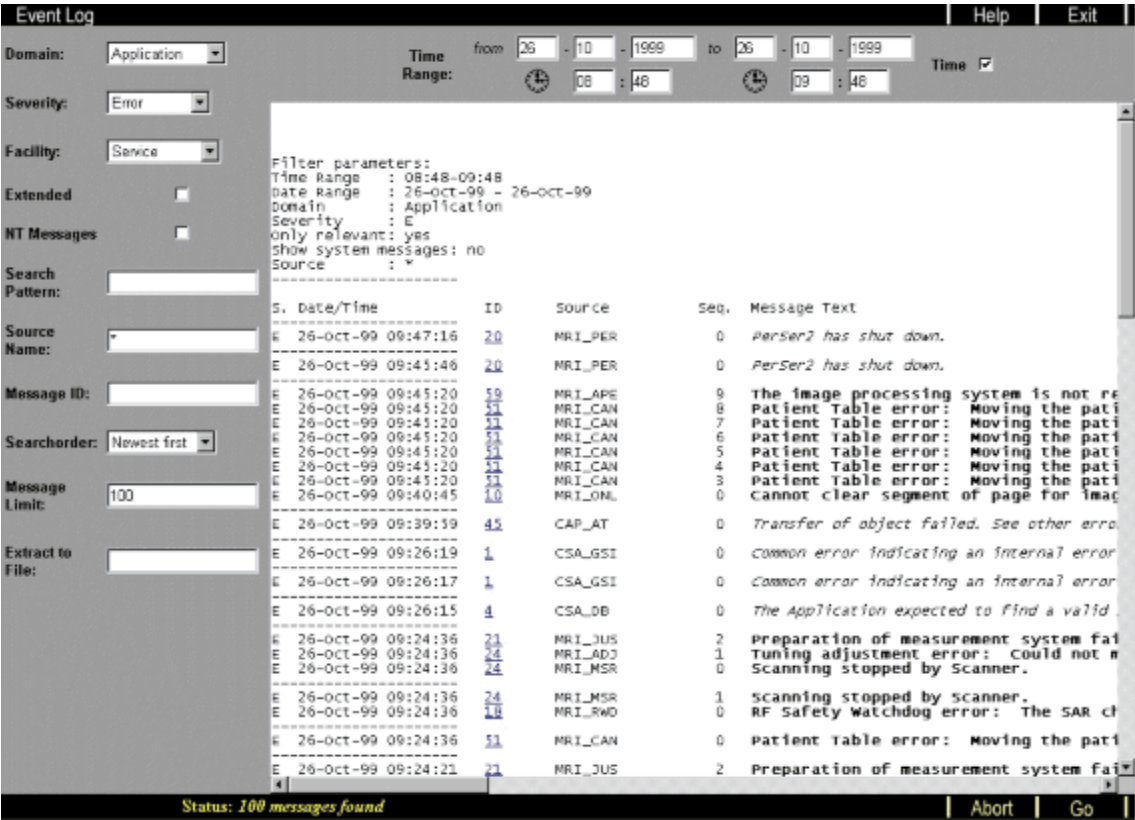
Fig. 1: Astex Beta-type Model 44 S26B - 123



1.6.2 Error Messages in the Event Log for RFPA 35 kW

In the **Event Log window**, available from the Service Software Homepage, you can select detailed information from the error logger. (→ Fig. 2 Page 14) shows the window with the possible selections.

Fig. 2: Event Log window



Each Message ID has a hypertext-link to an **Extended Error Message explanation** (→ Fig. 3 Page 15). Here you will find some more information in regard to the recommended action and an explanation of the error message.

Fig. 3: Example of an Extended Error Message

Message:	E	_287	_SUP_RFPK2180W35_1F
Text:	RF Power Amplifier error: Fault 38: PA FILAMENT UV. 1. System Standby -> System On 2. If the error persists call service		
Explanation:	1. PA Filament voltage is below specification or zero. 2. Defective PA Filament PS.		
Action:	Replace PS deck.		
Information:			
Last Update:	Tuesday, March 11, 2003		

- » The **Explanation** line gives you a more detailed description of the error.
- » The **Action** line recommends actions to solve the error.
- » Additional information may be found in the **information** line.

Tab. 1 Listing of error messages (RFPA 35 kW/ Tube Amp)

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Front panel LED's are blinking	NA	System Controller Fault	1. Re-initialise the amplifier via the MRI system software. 2. Call service.	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on the PS deck. Wait ten seconds. Then turn CB2 back ON. 2. If the fault persists replace the PS deck.
All front panel LED's are ON	NA	System Controller Fault	1. Re-initialise the amplifier via the MRI system software 2. Call service.	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on the PS deck. Wait ten seconds. Then turn CB2 back ON. 2. If the fault persists replace the PS deck.
Fault 00 No Fault	NA			
Fault 01 Phase Loss	Standby	1. Temporary or permanent loss of one of the primary power phases. 2. Open fuse(s)	1. Reboot scanner. 2. If the error persists call service.	1. If the Fault persists check voltage on each mains phase at input to amplifier. Correct if needed. 2. Check fuses F3-F5 and replace if needed. 3. Replace the Power supply.
Fault 02 SCR Overtemp	Standby	1. Blocked Airflow 2. Defective PS	1. Allow amplifier to cool for 20 minutes and then attempt to reboot the scanner. 2. Use a Siemens released sequence 3. If the error persists call service.	1. Check PS air filters and clean or replace if needed. 2. Verify that nothing is obstructing the air-flow from the front or the rear of the amplifier. 3. Replace the PS deck

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 03 HV Transformer Overtemp	Standby	1. Blocked Airflow 2. Defective PS	1. Allow amplifier to cool for 20 minutes and then attempt to reboot the scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Check PS air filters and clean or replace if needed. 2. Verify that nothing is obstructing the air-flow from the front or the rear of the amplifier. 3. Replace the PS deck.
Fault 04 LV Transformer Overtemp	Off	1. PS Fan(s) Failure or blocked air flow 2. Defective PS deck	1. Allow amplifier to cool for 20 minutes and then attempt to reboot the scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Check PS air filters and clean or replace if needed. 2. Verify that nothing is obstructing the air-flow from the front or the rear of the amplifier. 3. Replace the PS deck.
Fault 05 LV PS Overtemp	Off	1. Blocked airflow 2. Defective PS deck	1. Allow amplifier to cool for 20 minutes and then attempt to reboot the scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Check PS air filters and clean or replace if needed. 2. Verify that nothing is obstructing the air-flow from the front or the rear of the amplifier. 3. Replace the PS deck.
Fault 06 Spare				
Fault 07 Spare				
Fault 08 - 10 Not Used				

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 11 9V PS Low	Off	1. Defective CAN module 2. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. If Fault persists replace CAN Bus module. 2. If this doesn't clear the fault, replace the PS deck.
Fault 12 SSD +15V PS Low	Off	1. Defective RF deck	1. System standby then system on. 2. If the error persists call service.	1. Replace RF Deck.
Fault 13 SSD +15V PS High	Off	1. Defective RF deck	1. System standby then system on. 2. If the error persists call service.	1. Replace RF Deck
Fault 14 PA Bias Low	Off	1. Internal tube arc (this is common for oxide cathode tubes during the first hours of operation and after transit - this arc will not cause any damage) 2. Open PS fuse 3. Defective PA tube 4. Defective PS deck	1. System standby then system on. 2. Attempt restart three times. 3. If the error persists call service.	1. Check Fuse F9 - F11, F15 - F17 and replace if needed. 2. If the Fault persists replace the PA tube. 3. If a new tube does not correct the problem re-install the original tube and replace the PS deck.

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 15 PA HV Low	Standby	1. Incorrect Mains tap 2. Interconnect cable loose 3. Defective 48V Step PS 4. Defective PS deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the fault occurs at MRI start up and the system runs fine after the breaker is reset, contact service. 4. If the error persists call service.	1. Check the Mains tap (only at installation!). 2. Verify that the 4-pin barrel interconnect cable between the RF and PS deck is secure. 3. If the fault persists replace the 48V Step PS by performing update instruction S1032/05/I. 4. If the fault persists replace the PS deck.
Fault 16 IPA HV Low	Standby	1. Incorrect Mains tap 2. Interconnect cable loose 3. Defective 48V Step PS 4. Defective PS deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the fault occurs at MRI start up and the system runs fine after the breaker is reset, contact service. 4. If the error persists call service.	1. Check Mains tap (only at installation!). 2. Verify that the 4-pin barrel interconnect cable between the RF and PS deck is secure. 3. If the fault persists replace the 48V Step PS by performing update instruction S1032/05/I. 4. If the fault persists replace the PS deck.
Fault 17 Not Used				
Fault 18 SSD 5 V PS Low	Off	1. Defective RF deck	1. System standby then system on. 2. If the error persists call service.	1. If the fault persists replace the RF deck
Fault 19 SSD 5V PS High	Off	1. Defective RF deck	1. System standby then system on. 2. If the error persists call service.	1. If the fault persists replace the RF Deck.

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 20 - 22 Not Used				
Fault 23 9V PS High	Off	1. Defective RF deck	1. System standby then system on. 2. If the error persists call service.	1. If the fault persists replace the PS Deck.
Fault 24 SSD -15V PS Low	Off	1. Defective RF deck	1. System standby then system on. 2. If the error persists call service.	1. If the fault persists replace the RF Deck.
Fault 25 SSD -15V PS High	Off	1. Defective RF deck	1. System standby then system on. 2. If the error persists call service.	1. Check fuses F6-F8 and replace if needed. 2. If the fault persists replace the RF Deck.
Fault 26 IPA Bias Low	Off	1. Internal tube arc (this is common for oxide cathode tubes during the first hours of operation and after transit - this arc will not cause any damage) 2. Open PS fuse 3. Defective IPA tube 4. Defective PS deck	1. System standby then system on. 2. Attempt restart three times. If the error persists call service.	1. Check Fuse F9-F10, F15-F17 and replace if needed. 2. If the Fault persists replace the IPA tube. 3. If a new tube does not correct the problem re-install the original tube and replace the PS deck.
Fault 27 PA HV High	Standby	1. Incorrect Mains tap 2. Defective PS deck	1. Reboot scanner. 2. If the error persists call service.	1. Check Mains tap (only at installation or replacement!). 2. If the fault persists replace the PS deck.

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 28 IPA HV High	Standby	1. Incorrect Mains tap 2. Defective PS deck	1. Reboot scanner. 2. If the error persists call service.	1. Check Mains tap (only at installation or replacement!). 2. If the fault persists replace the PS deck.
Fault 29 PA Plate Current (Ip) High	Standby	1. Incorrect load or transmit path problem 2. Excessive RF drive level for load 3. Defective RF deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Confirm system Tx/Rx circuitry is functioning properly. (Use SeSo Test Tools). 2. Verify that the amplifier is terminated into a suitable RF load. 3. Replace the RF deck.
Fault 30 IPA Plate Current (Ip) High	Standby	1. Incorrect load or transmit path problem 2. Excessive RF drive level for load 3. Defective RF deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Confirm system Tx/Rx circuitry is functioning properly. (Use SeSo Test Tools). 2. Verify that the amplifier is terminated into a suitable RF load. 3. Replace the RF deck.
Fault 31 PA Grid Current (Ig) High	Standby	1. Incorrect load or transmit path problem 2. Excessive RF drive level for load 3. Defective RF deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Confirm system Tx/Rx circuitry is functioning properly. (Use SeSo Test Tools). 2. Verify that the amplifier is terminated into a suitable RF load. 3. Replace the RF deck.
Fault 32 IPA Grid Current (Ig) High	Standby	1. Incorrect load or transmit path problem 2. Excessive RF drive level for load 3. Defective RF deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Confirm system Tx/Rx circuitry is functioning properly. (Use SeSo Test Tools). 2. Verify that the amplifier is terminated into a suitable RF load. 3. Replace the RF deck.

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 33 - 34 Not Used				
Fault 35 50V PS Low	Off	1. Open PS fuse(s) 2. Defective PS Deck 3. Defective RF deck	1. System standby then system on. 2. If the error persists call service.	1. Check Fuse F15, F16, F17. 2. Remove the right side panel of the RF deck. Remove the D-sub connection labeled "SSD". While starting the warm up sequence monitor the DC voltage between the two pins of the connector. 3. If you measure 60 - 64 VDC prior to the fault, replace the RF deck. 4. If no voltage is present, replace the PS deck.
Fault 36 50V PS High	Off	1. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. If the fault persists replace the PS deck.
Fault 37 PA Filament Over Voltage	Off	1. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. Replace PS deck.
Fault 38 PA Filament Under Voltage	Off	1. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. Replace PS deck.
Fault 39 IPA Filament OV	Off	1. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. Replace PS deck.

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 40 IPA Filament UV	Off	1. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. Replace PS deck.
Fault 41 PA Air Flow	Off	1. Blocked airflow in RF deck 2. Loose deck interconnect cable 3. Open fuse in the amplifier 4. Defective RF deck	1. System standby then system on. 2. If the error persists call service	1. Check RF deck air filters and clean or replace if needed. 2. Verify that nothing is obstructing the airflow from the front or the rear of the amplifier. 3. Verify that the interconnect cable P5 between the RF and PS deck is secure. 4. Check fuses F1 and F2 and replace if needed. Replace the RF Deck.
Fault 42 Logic Shutdown (RF_PA_ON)	Standby	1. Interface connection problem 2. Defective CAN Bus module 3. Defective PS deck	1. Check to determine if the MRI system is in a state that allows the amplifier to be enabled. 2. Reboot scanner. If the error persists call service.	1. Check/replace Fiber Optic command from CCA/MMC/D27/U1 to amplifier RPFA_ON for proper connection and operation. 2. Replace CAN Module. 3. If fault persists, contact HSC Erlangen, before replacing the PS Deck.
Fault 43 Not Used				
Fault 44 DSP Failure	Off	1. Disruption in system power 2. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on the PS. Wait ten seconds. Then turn CB2 back ON. 2. If the fault persists replace the PS Deck

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 45 PS Cover Open	Off	1. One of the PS deck access covers is off or loose.	1. System standby then system on 2. If the error persists call service.	1. Verify that all PS access covers are in place and tight. 2. Check function of switches.
Fault 46 RF Deck Cover Open	Off	1. One of the RF deck access covers is off or loose. 2. Loose deck interconnect cable.	1. System standby then system on 2. If the error persists call service.	1. Verify all RF deck access covers are in place and tight. 2. Verify that P1 interconnect cable between the RF and PS deck is secure.
Fault 47 PA Air too Hot	Standby	1. Selected RF pulse sequence exceeds amplifier power dissipation capability. 2. Blocked PA air exhaust 3. Equipment room ambient is > 40 degrees C. 4. Defective temperature probe 5. Defective RF deck	1. Allow amplifier to cool for 20 minutes and attempt then reboot scanner. 2. Use a Siemens released sequence. If the error persists call service.	1. Check RF deck air filters and clean or replace if needed. 2. Ensure equipment room ambient is < 40 degrees C. 3. Replace the temperature probe. 4. If the fault persists replace the RF deck.
Fault 48 PA Air Temperature Monitor	Off	1. Blocked PA air flow 2. Equipment room ambient is > 40 degrees C. 3. Defective RF deck	1. System standby then system on. 2. If the error persists call service.	1. Check RF deck air filter and clean or replace if needed. 2. Ensure equipment room ambient is < 40 degrees C. 3. If the fault persists replace the RF Deck.
Fault 49 Not Used				

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 50 Non-Fatal Fault in Standby	Off	1. Disruption in system power 2. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on the PS. Wait ten seconds. Then turn CB2 back ON. 2. System Standby then system on 3. If the fault persists replace the PS deck.
Fault 51 - 57 Not Used				
Fault 58 SSD Driver Temp High	Standby	1. Selected RF pulse sequence may exceed amplifier capabilities 2. Blocked airflow 3. Equipment room ambient is > 40 degrees C. 4. Defective RF deck	1. Allow amplifier to cool for 20 minutes and attempt then reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Check RF deck air filters and clean or replace if needed. 2. Ensure the equipment room ambient is < 40 degrees C. 3. If fault persists replace the RF Deck.
Fault 59 SSD Preamp Temp High	Standby	1. Selected RF pulse sequence may exceed amplifier capabilities. 2. Blocked airflow 3. Equipment room ambient is > 40 degrees C. 4. Defective RF deck	1. Allow amplifier to cool for 20 minutes and attempt then reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Check RF deck air filters and clean or replace if needed. 2. Ensure the equipment room ambient is < 40 degrees C. 3. If fault persists replace the RF Deck.
Fault 60 Not Used				

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 61 PA Bias Current	Standby	1. Open fuse 2. PA tube failure or end-of-life 3. Defective PS deck	1. Reboot scanner. 2. If the error persists call service.	1. Check F18-F20 and replace if needed. 2. Replace PA tube. 3. If a new tube does not correct the problem re-install the original tube and replace the PS deck
Fault 62 IPA Bias Current	Standby	1. Open fuse 2. IPA tube failure or end-of-life 3. Defective PS deck	1. Reboot scanner. 2. If the error persists call service.	1. Check fuses F21-F23. 2. Replace IPA tube. 3. If a new tube does not correct the problem re-install the original tube and replace the PS deck.
Fault 63 NVRAM Failure	Off	1. Disruption in system power 2. Defective PS deck	1. System standby then system on. 2. If the error persists call service	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on the PS. Wait ten seconds. Then turn CB2 back ON. 2. System Standby then system on. 3. If the fault persists replace the PS Deck.
Fault 64 PMU Failure	Off	1. Disruption in system power 2. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. Re-initialize the amplifier by turning OFF CB2 on the PS. Wait ten seconds. Then turn the PS circuit breaker back ON. 2. System Standby then system on. 3. If the fault persists replace the PS Deck.
Fault 65 Not Used				
Fault 66 SSD EPROM Failure	Off	1. Disruption in system power 2. Defective RF deck	1. System standby then system on. 2. If the error persists call service.	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on the PS. Wait ten seconds. Then turn CB2 back ON. 2. System Standby then system on 3. Verify that P3 and P4 interconnect cables between the RF and PS decks are secure.

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 67 IPA Tube Arc	Standby	1. Internal tube arc (this is common for oxide cathode tubes during the first hours of operation and after transit - the arc will not cause any damage) 2. Defective RF deck	1. System standby then system on 2. Attempt restart three times. 3. If the error persists call service.	1. If the Fault persists replace the IPA tube. 2. If a new tube does not correct the problem re-install the original tube and replace the RF deck.
Fault 68 PA Tube Arc	Standby	1. Internal tube arc (this is common for oxide cathode tubes during the first hours of operation and after transit - the arc will not cause any damage) 2. Defective RF deck	1. System standby then system on. 2. Attempt restart three times. 3. If the error persists call service.	1. If the Fault persists replace the PA tube. 2. If a new tube does not correct the problem re-install the original tube and replace the RF deck.
Fault 69 - 70 Not Used				
Fault 71 SSD Overdrive	Standby	1. RF input level is too high. 2. Defective RF deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Check the RF input level of Amplifier. 2. If the fault persists replace the RF Deck.
Fault 72 Not Used				

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 73 SSD Reflected Power	Standby	1. Defective IPA tube 2. Defective RF deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. If the Fault persists replace the IPA tube. 2. If a new tube does not correct the problem re-install the original tube and replace the RF deck
Fault 74 DSP Comm. Failure	Off	1. Disruption in system power 2. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on the PS. Wait ten seconds. Then turn CB2 back ON. 2. System Standby then system on. 3. If the fault persists replace the PS deck.
Fault 75 DSP Interrupt	Off	1. Disruption in system power 2. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on the PS. Wait ten seconds. Then turn CB2 back ON. 2. System Standby then system on. 3. If the fault persists replace the PS deck.
Fault 76 - 86 Not Used				
Fault 87 IPA Plate Over Current	Standby	1. RF input level too high 2. High load or "no-load" 3. Defective IPA tube 4. Defective PA tube	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Confirm system Tx/Rx circuitry is functioning properly. (Use SeSo Test Tools). 2. If fault persists replace the IPA tube. If fault continues re-install the original IPA tube, then replace the PA tube. 3. If a new tube does not correct the problem re-install the original tube and replace the amplifier.

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 88 PA Plate Over Current	Standby	1. RF input level too high 2. High load or "no-load". 3. Defective PA tube	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Confirm system Tx/Rx circuitry is functioning properly. (Use SeSo Test Tools). 2. If fault persists replace the PA tube. 3. If a new tube does not correct the problem re-install the original tube and replace the amplifier.
Fault 89 - 92 Not Used				
Fault 93 Un-blank Exceeded Limit	Standby	1. Amplifier un-blanked for > 256 msec 2. Amplifier un-blank duty cycle exceeds 50% mode High, or 99% mode Low. 3. Defective CAN Bus module 4. Defective PS deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Verify that an approved RF sequence is being used. 2. If fault persists replace the CAN Bus module. 3. If this doesn't correct the fault reinstall the original CAN Bus module then replace the PS deck.
Fault 94 RF Burst Exceeded Limit	Standby	1. The RF average power over a 300 msec period exceeded 3,500 watts in mode High or 1,600 watts in mode Low. 2. Defective RF deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. If fault persists replace the RF Deck.

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 95 Forward Power High	Standby	1. The RF peak power exceeded 37,500 watts in mode High or 8,500 watts in mode Low. 2. High load or "no-load" 3. Defective RF deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Confirm system Tx/Rx circuitry is functioning properly. (Use SeSo Test Tools). 2. Verify that the RF drive level to the amplifier is accurate. 3. Verify amplifier calibrations (QA/ RF Verify). 4. If fault persists replace the RF Deck.
Fault 96 Reflected Power High	Standby	1. High load or "no-load" 2. Excessive RF drive level for load 3. Defective RF deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Confirm system Tx/Rx circuitry is functioning properly. (Use SeSo Test Tools). 2. Check to make sure the RF output cable is secure on the amplifier. 3. Verify that the amplifier is terminated into a suitable RF load. 4. If fault persists replace the RF deck.
Fault 97 Average Power Trip	Standby	1. The RF average power over a three second period exceeded 1,750 watts in mode high or 800 watts in mode Low. 2. Defective RF Deck	1. Reboot scanner. 2. Use a Siemens released sequence. 3. If the error persists call service.	1. Verify that an approved RF sequence is being used. 2. If fault persists replace the RF deck

Fault Description	State after Fault	Probable Cause	Customer Action	Service Action
Fault 98 Loss of Power	Off	1. Mains power to the amplifier was interrupted or one of the mains circuit breakers on the amplifier were turned "Off". 2. Defective PS deck.	1. System standby then system on. 2. If the error persists call service.	1. Verify that mains power is present and circuit breakers CB1 and CB2 are ON. 2. If fault persists replace the PS deck
Fault 99 Undefined	Off	1. Defective PS deck	1. System standby then system on. 2. If the error persists call service.	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on the PS. Wait ten seconds. Then turn CB2 back ON. 2. If fault persists replace the PS deck

The following table shows general system performance issues which will cause no specific or no error messages at all.

Tab. 2 System performance issues (No Fault Code in the Event log)

Symptom	Possible Cause	Service Action
Front panel LEDs are blinking	■ Disruption in system power ■ Defective PS deck	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on the PS deck. Then turn CB2 back ON. 2. If the fault persists, replace the PS deck.
Forward and reflected power monitor issues (DICO monitor)	1. Loose or damaged interface connections 2. Loose deck interconnect cable 3. Defective RF deck	1. Verify that FRX5 and FRX6 monitor connections on the front of amplifier have no damage and are secured. Correct if needed. 2. Verify that the deck interconnect cable P6 is secured. 3. If the fault persists, replace the RF deck.
No power to amplifier	1. Amplifier circuit breaker(s) are off 2. Loss of AC mains voltage. 3. Open fuse in the PS deck 4. Defective PS deck	1. Verify that CB1 and CB2 breakers are on. 2. Verify that AC voltage is present to amplifier 3. Check fuses F12 - F15 and replace if needed. 4. If the problem persists, replace the PS deck.

Symptom	Possible Cause	Service Action
RF characteristic issues: Unstable output, RF noise or spikes	1. Loose RF output cable 2. Loose PA tube clamp 3. Defective RF deck	1. Verify that the RF output cable is secured and that there are no signs of arcing at the connector. Correct if needed. 2. Verify that the PA tube clamp is properly installed with no loose hardware. 3. If the fault persists, replace the RF deck.
All front panel LEDs are ON.	System controller fault	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on the PS deck. Wait ten seconds. Then turn CB2 back ON. 2. If the fault persists, replace the PS deck.
No RF power at output and no fault indication	1. Defective FOCs 2. Wrong or FOCs are not connected. 3. Defective CAN module 4. Defective PCI-MON	1. Check FOCs and functionality of connections. 2. Run TestTools/AMC tests. 3. Replace the CAN module of the RFPA.

1.7 Error Messages

The error messages are listed in the (→ SKB, <https://skb.healthcare.siemens.com/>).

2.1 General

This section describes additional tests for troubleshooting on components (FRU's) of the RF-System, such as voltage level and pulse form measurements at RFSU (modulator, synthesizer), RFPA, RFCL and local coil connectors.

2.1.1 Required Tools

- Function Description RF system (FUN)
- Diagrams RF system
- Set of standard tools
- 50 Ohm/2W BNC Termination resistor (7760457 B1947)
- QLA T-connector (e.g. Suhner 43QLA 01-0-1, or equivalent)
- Service cables QLA-BNC (2× 8733446 K2060)
- Connector BNC-Banana plug (3025749 B1947)
- Digital multimeter, type Fluke 8060A or equivalent
- Oscilloscope, 200MHz bandwidth
- Service plug for LCCS (4795501 for Harmony, Symphony and Sonata; 7131050 for Trio).



Please ensure that the system is powered on at least for 20 minutes before performing any measurements and/or running any functional or interactive test of the Service Software procedures (TestTools/RF Tests), to stabilize the temperature in the control cabinet.

2.2 RFSU

Measurement of:

- Modulator SSB1 signal
- RFPA PF (forward wave) signal
- Synthesizer 10 MHz signal
- TALES CV signal (with scope)



If the MR system is equipped with a second RX Module, these modules can be swapped for troubleshooting.

2.2.1 SSB1-Signal

2.2.1.1 Prerequisites

(Measurement of SSB1 and PF/RFAS)

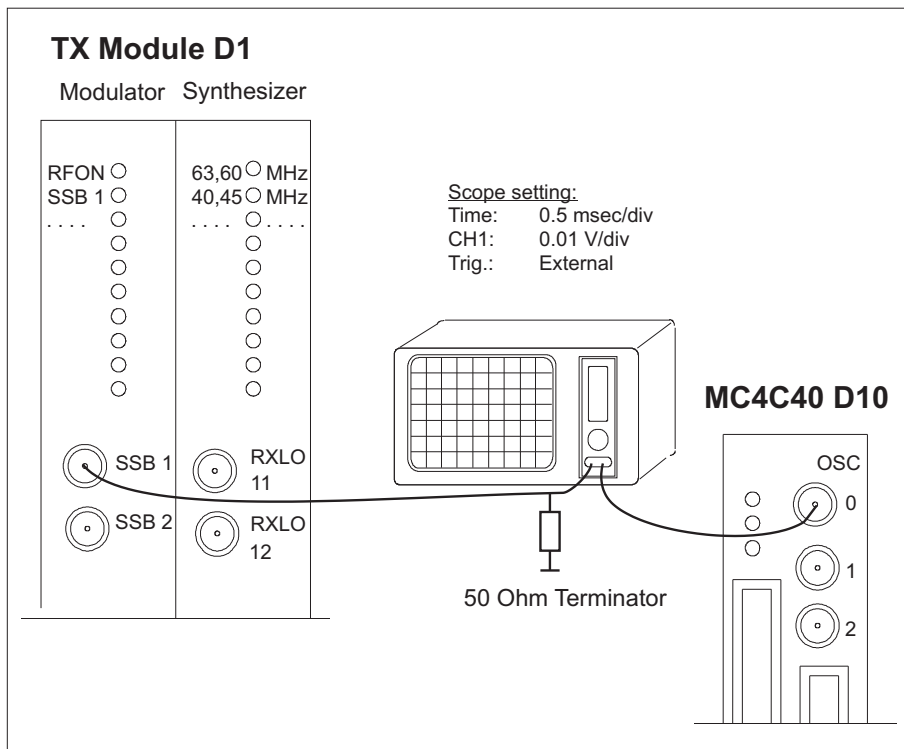
- MR system ready for measurement
- Tuned body phantom with loader positioned in the magnet center.

2.2.1.2 Procedure

- Start the Service Software platform.
Run QA/Spike Check. The Service patient will be used.
- Reduce the Service platform to an icon.
- From the examination manager
 - Siemens/Sequence Region/Service Sequences/ Default Protocols -
fetch the sequence rf_loop.
- Open the sequence. Adjust the following parameters:
Click on System and Transmitter/Receiver:
Set RF-amplitudes sSRF01 - sSRF06 to 100 V.
Click on Contrast:
Set Measurements to 100.

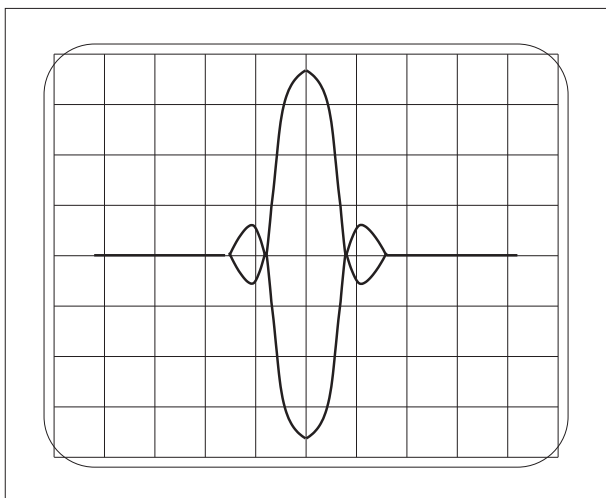
- Start the measurement by clicking on Apply.

Fig. 4: Measurement configuration - Signal modulator SSB1



- Oscilloscope setting: CH1: 0.01 V/div, Time: 0.5 ms/div, ext.Trigger.
- Monitor the SSB1 signal (first pulse = constant sinc-pulse) on the oscilloscope.
- A symmetrically-shaped signal must be continuously present, without voltage drop-off or overshoots.

Fig. 5: Typical signal SSB1 (first sinc-pulse of sequence rf_loop)



- Abort/Stop the measurement
- Restore the original system configuration.

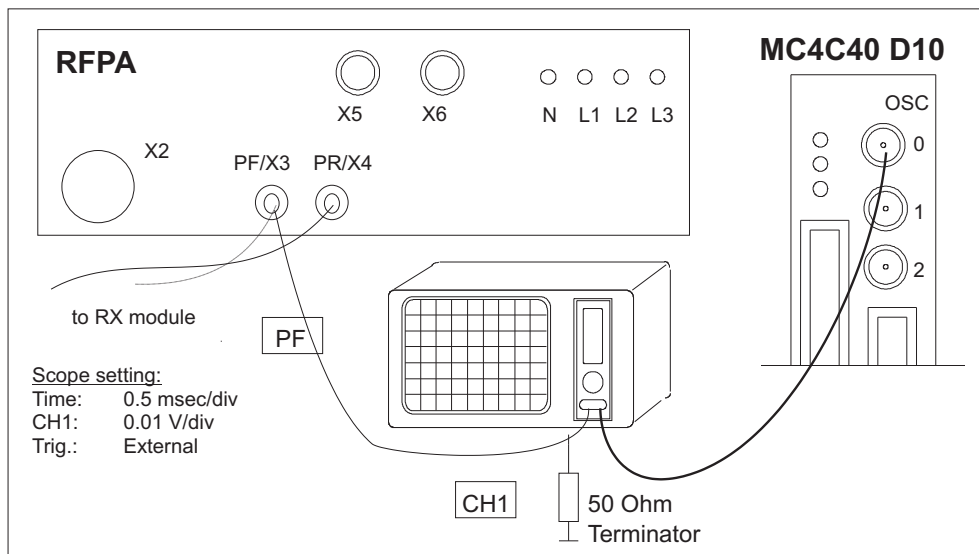
2.3 RFPA

2.3.1 Directional Coupler Signal PF

2.3.1.1 Procedure

- Remove the cable to the RX module from PF output of RFPA. Connect the CH1 to X3/PF (forward wave of the directional coupler of RFPA) via 50 Ohm terminator. Connect the MC4C40 output OSC 0 as an external trigger to the oscilloscope.
- Start the sequence. The sequence is the same as for the SSB1 signal test.

Fig. 6: Measurement configuration - RFPA signal PF/PR

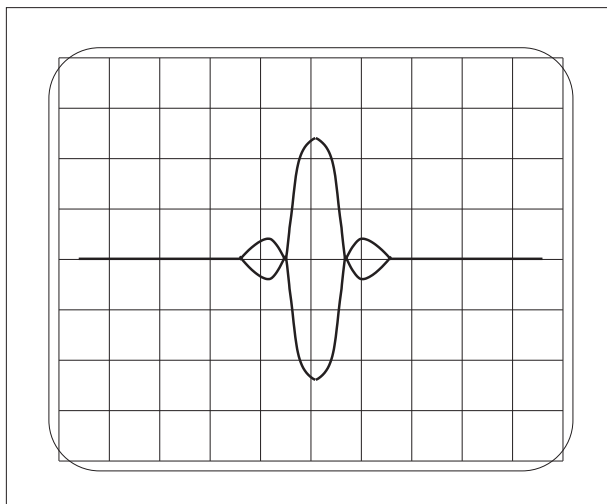


- Oscilloscope setting: CH1: 0.01 V/div, Time: 0.5 ms/div, ext.Trigger.

- Monitor the signal PF (first pulse = constant sinc-pulse) on the oscilloscope: A symmetrically-shaped signal must be continuously present, without voltage drop-off or overshoots. Refer to (→ 1, 3/ Fig. 7 Page 38).

The attenuation of the directional coupler is approx. 70 dB at 63 MHz.

Fig. 7: Typical signal RF/RFA (first sinc-pulse of sequence rf_loop)



- Abort the measurement
- Restore the original system configuration.

2.3.2 Tripped-off Circuit Breaker F11 (DORA/ PORA)



CAUTION

The RFPA can be seriously damaged by improper power switching directly at circuit breaker F11/ACC !

Switching-on the RFPA supply voltage directly at circuit breaker F11/ACC can cause irreversible damage to the RFPA soft-start unit!

- » Before operating circuit breaker F11/ACC, always first switch-off the MR system to stand-by.
- » Then, wait at least 3 minutes before switching circuit breaker F11/ACC.
- » Never switch power to the RFPA directly at circuit breaker F11/ACC !

Correct RFPA power switch-on sequence (after circuit breaker F11/ACC has tripped-off):

- Switch the MR system to **Standby** at the Alarm Box.
- Wait at least 3 minutes to discharge RFPA voltages..
- Switch-on RFPA circuit breaker F11/ ACC..
- Switch the MR system **ON** at the Alarm Box.
- **Investigate the cause**, measure the RFPA supply voltages as follows:

Tab. 3 AC supply voltage at DORA/ PORA

Point of measurement	L1- L2	L2- L3	L3- L1	Maximum charged deviation (max-min)
Spec	166,3 V +/- 8 V			0,8 V
Result				

Point of measurement	L1-N	L2-N	L3-N	Maximum charged deviation (max- min)
Spec	96 V +/- 4 V			0,5 V
Result				

Tab. 4 Testing range

Point of measurement	Spec	Measurement value	Testing range	
N- PE	117 V +/- 2 V		DC	
	17 V +/- 2 V		AC	



Voltages are measured at the specified AC line voltage of the MR system (see Planning Guide).

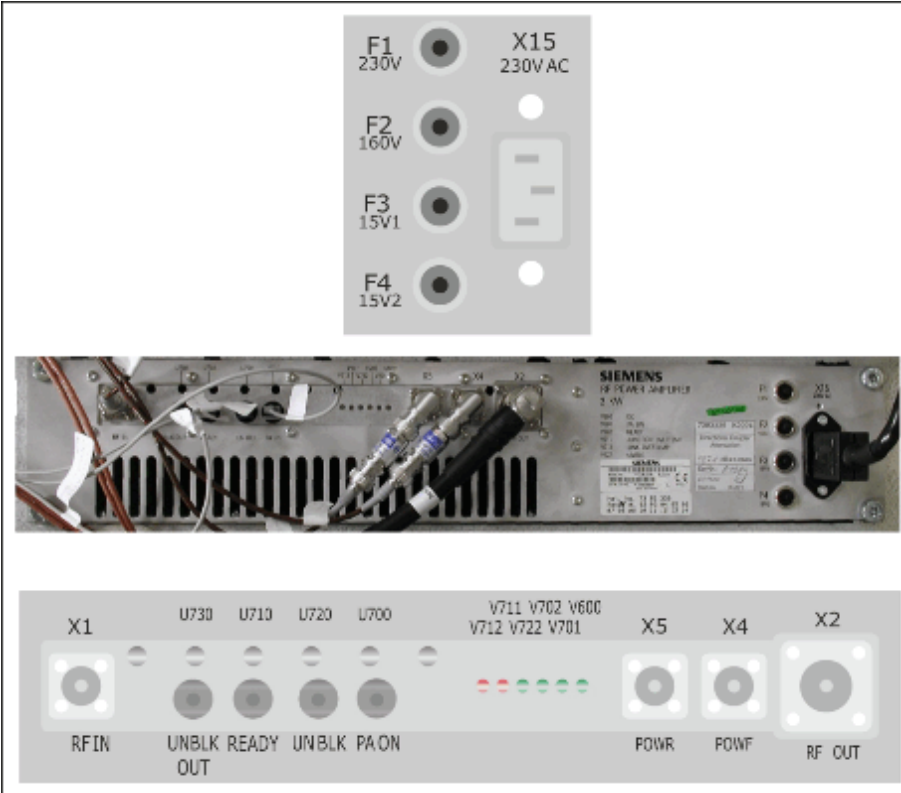
Line voltage fluctuations can cause deviations from the expected results.

2.3.3 CORA Shut-downs



The CORA RFPA (material number 7383339) is part of the Multinuclear Spectroscopy option.

Fig. 8: CORA RFPA front view



In case the status of the CORA RFPA is not ok (e.g. amplifier shutdowns for unknown reason), perform the following checks:

- Check if the three green LEDs (V600, V701, V702) at the CORA are **ON** and green LED V722 is flashing when a sequence is running:

Tab. 5 LEDs at CORA

LED	Description
V600 "DC" (green)	ON indicates internal power supply unit OK
V701 "PA ON" (green)	ON indicates amplifier is being enabled (no RFSWD errors). The PA_ON signal comes from IOP/ LWL_INT board via FOC connector U700 at CORA.
V702 "READY" (green)	ON indicates amplifier start-up completed and unit now ready.
V722 "UNBLK" (green)	ON (flashing) indicates unit being unblanked (i.e., when transmitting).

LED	Description
V711 "JUNCTION OVERTEMP" (red)	ON error indicates transistor junction temperature too high (non-latched).
V712 "SINK OVERTEMP" (red)	ON error indicates heat sink temperature too high (non-latched).

- In case a **CORA shuts-down already at sequence start**, check:
 - if the max. pulse amplitude is too high
 - if the RF-pulse duration is too long (wrong sequence parameters, customer-programmed sequence?)
 - for correct timing of UNBLANK signal. Swap FOC "UNBLK" CORA with DORA. If the error at CORA disappears, replace "UNBLANK" FOC cable from IOP/ LWL_INT board.
 - for interrupted PA_ON signal at FOC U700 (broken FOC?, IOP/ LWL_INT defect?)
 - remove the CORA RFPA input (X1, RF_IN) and start sequence again. If the error persists, the shut-down is NOT caused by too high max. pulse amplitude but may result from a too long pulse duration or interrupted PA_ON signal at FOC U700.



CORA shut-downs at sequence start are usually NOT caused by RFPA over temperature and are NOT caused by too high average power.

- In case a **CORA shuts-down after some measurement time**, check:
 - the RFPA cooling water temperature and flow under **SeSo > Magnet&Cooling > Cooling Status**.
 - if the average power demanded by the sequence is too high (wrong sequence parameters, customer-programmed sequence?)

2.4 RFPA1/2 (Trio)

2.4.1 Directional Coupler Signal PF of RFPA1/2 (RFPA 35 kW or 8 kW)

2.4.1.1 Prerequisites

- MR system ready for measurement
- Body phantom with loader (RFPA 35 kW) or Head Coil with bottle phantom (RFPA 8 kW) positioned in the magnet isocenter.

2.4.1.2 Performing the Procedure

- Remove the QLA connector from input PF(RFPA 35 kW or 8 kW) of receiver #1 or #3. Connect a QLA T connector to input PF input of receiver #1 or #3. Connect the cable from PF output of (RFPA 35 kW or 8 kW) (forward power directional coupler of RFPA) at one end of the T and the scope CH1 via feedthrough 50 Ohm terminator to the other end. Connect the AMC/PCI-TX/X8 output OSZI_TRIG (0) as an external trigger to the scope.
- Use the same service sequence as for the SSB1 signal test.
- Oscilloscope setting: CH1: 0.01 V/div, Time: 0.5 ms/div, ext.Trigger.
- Monitor the signal PF (constant sinc-pulse) on the oscilloscope.
 - » A symmetrically-shaped sinc signal must be continuously present without voltage drop-off or overshoots.
- Abort the measurement.
- Restore the original system configuration.

2.5 RFPA 35 kW Trio (Tube Amp)

2.5.1 General Safety Information for RFPA 35kW



WARNING

High voltage!

Hazard of electric shock or burn.

Even when the MR system is switched off, mains voltage at the mains input of the amplifier is present.

Therefore:

Before any work on the amplifier switch off the circuit breakers CB2 and then CB1 at the amplifier and at least circuit breaker F25 in the mains box (CCA) and ensure against accidental switch-on.

- » Observe the warning information on the amplifier.
- » After the amplifier has been turned off, allow at least ten minutes for the high voltage filter capacitors to discharge fully before removing any of the cover plates.

2.5.1.1 Meaning of Front Panel Status LED's

Status	Probable Cause	Action
All LED's are blinking	System controller error	1. Re-initialize the amplifier by turning OFF circuit breaker CB2 on PS deck. Wait ten seconds. Then turn the PS circuit breaker back ON. 2. If the fault persists replace the PS deck.
All LED's are ON		

FOC's to/from RFPA

FOC	Meaning	Status
U2/3	CAN1	LED's TX/RX on during bus activity
U4/5	CAN2	
U7	ENABLE-OUT	Light on, if Amplifier goes from Stdbby to Operate; the signal is then sent to RFPA 8 kW and enables it.
U6	RFPA-ON	Light must be on when the command "OPERATE" (CAN-Bus) switches on the high voltage for the tubes. (Transition STANDBY -> OPERATE)

FOC	Meaning	Status
U1	RFPA-UNBLANK	Amplifier is UNBLANKED if light is on
U8	STANDBY-OFF	Signal from D27 at MMC backplane. Tube filament will be switched on/off if light is on/off. (Transition Off <-> Stdby)

2.5.2 10 MHz-Signal

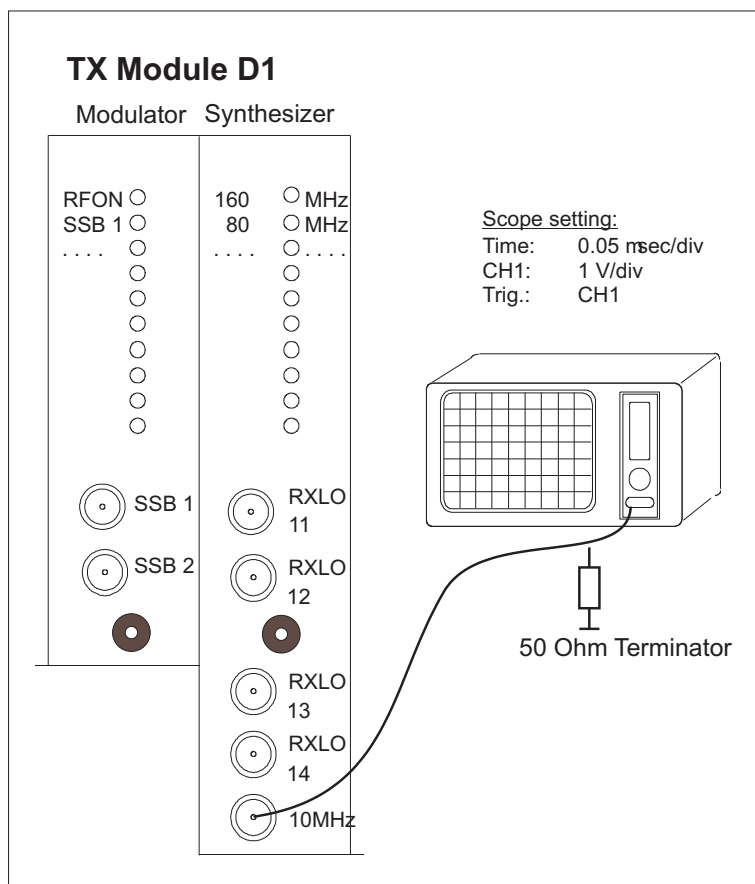
2.5.2.1 Prerequisite

- MR system is on.

2.5.2.2 Procedure

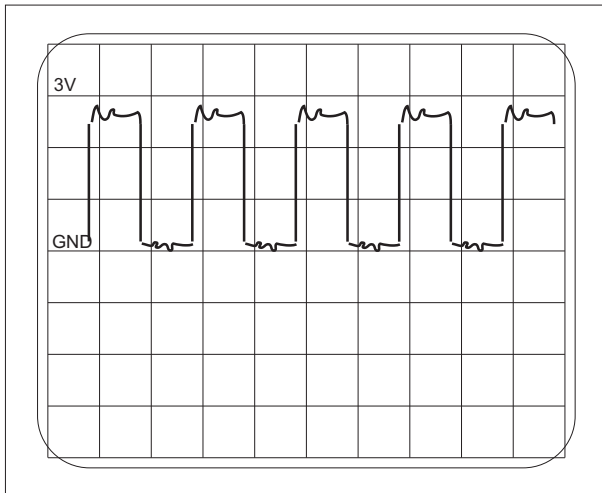
- Connect the oscilloscope to synthesizer output 10MHz via 50 Ohm terminator.
- Oscilloscope setting: CH1: 1V/div, Time: 0.05 μ sec/div, Trigger CH1.

Fig. 9: Measurement configuration - Synthesizer 10 MHz signal



- Monitor the 10 MHz signal on the oscilloscope (TTL-signal)
Nominal value:
High level: 2.6 V +/- 0.2V into 50 Ohm
Low level: max. 0.55 V into 50 Ohm (rectangular shaped).

Fig. 10: Typical signal 10 MHz



- Restore the original system configuration.

2.6 RFAS

2.6.1 500 mV Reference Voltage

2.6.1.1 Prerequisite

MR system is on.

2.6.1.2 Procedure

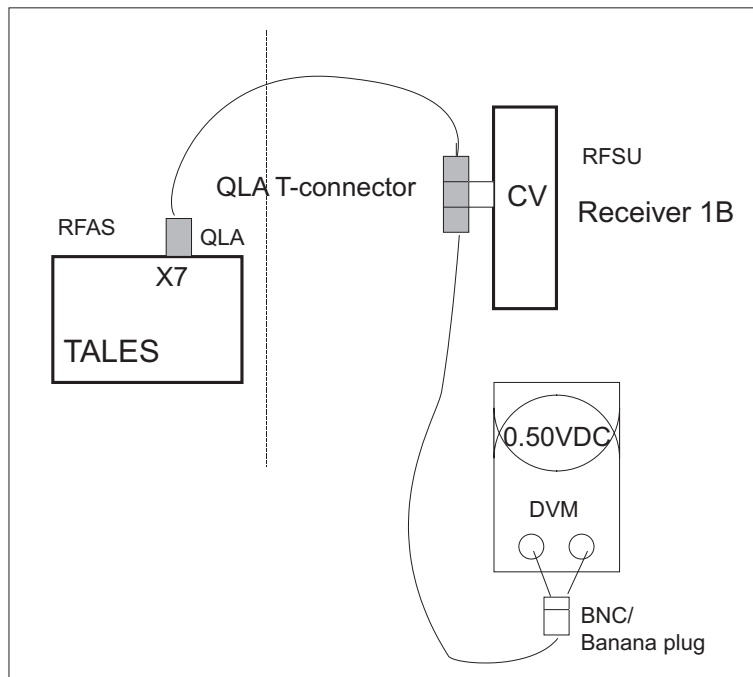
- Connect the cable from X7/TALES and a DVM via QLA T-connector to the CV input of Receiver 1/B.
- Switch off the circuit breaker F20 (main power supply unit for MMC) in the CCA cabinet.
- Switch on F20 again
 - » During the self test of the MMC (duration approx. 1 minute, red LED's at RX/TX modules "TEST" light up), the reference voltage of TALES can be measured with the DVM:
Nominal voltage: 0.475 to 0.525 VDC.

2.6.1.3 Additional Test of output Multiplexer control (TALES)

- Remove the three FOC's U1,2,3 from TALES in the idle state of the MR system.
 - » All three FOC's are dark
- Switch off/on F20.
 - » During the self test of MMC, all three FOC's light up.

(=> the reference voltage is switched to output X7/TALES by the multiplexer control).

Fig. 11: Measurement configuration - TALES Reference voltage



- Restore the original system configuration.

2.6.2 TALES CV-Signal

2.6.2.1 Measurement of the CV Signal with a scope

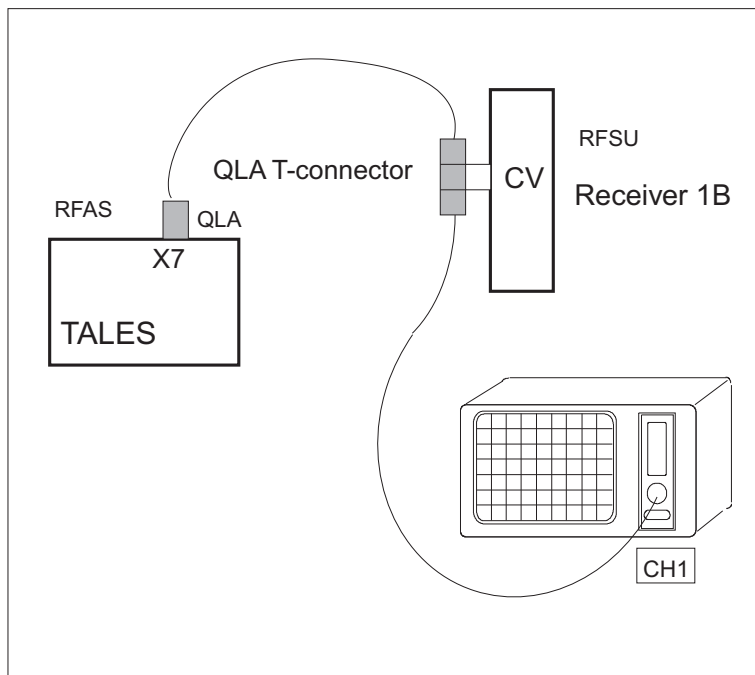
- Checking the correct function of TALES.
- Identifying additional RF interferences and errors in the TALES-Receiver line (CV input).
- Checking the correct connection of FOC control lines MH_SEL U1, 2, 3 at TALES.

2.6.2.2 Measurement Configuration

- Using a QLA T-connector, connect the CV signal from TALES and the scope with RFSU/ Receiver 1B as shown below.

- Set scope CH1 to 0.1 mv/div and time scale to 2 μ s/div.

Fig. 12: Measurement configuration - CV Signal



- Position the body phantom and loader in the magnet; register a suitable test patient.
- Load and open the service sequence **rf_pulse**:
Select/check the sequence parameter cards:
 - Resolution: Base resolution: **128**
 - Contrast: TR = 101ms, TE = 1ms, TI = 7
Measurements **20** (4:19 min measurement time)
- System, Transmitter/Receiver, Transmitter amplitude 200V (the pulse amplitude must be entered for each new measurement)
- Start the measurement. The following examples of waveforms correspond to a normal working TALES and intact signal line from TALES to Receiver B CV input.
- Perform the measurement with different transmitter amplitudes and check the corresponding waveform changes on the scope.

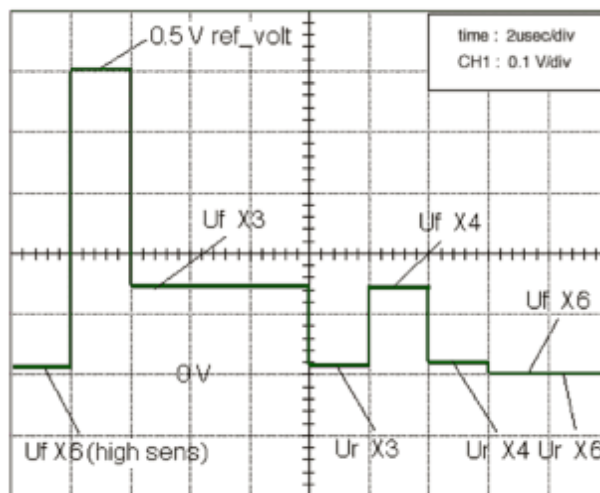
2.6.2.3 Waveform Display on scope

Under the measurement conditions described, the chronological sequence of the waveform segments is always the same. However, the display on the scope, i.e. starting point, depends on the trigger conditions. As a point of orientation, use the 0.5V reference voltage segment.

The signal heights of forward and reflected voltages are proportional to the transmitter voltage used.

MR system Symphony (TX to BC; transmitting to BC)

Fig. 13: Transmitter amplitude 200V, transmitting to body coil



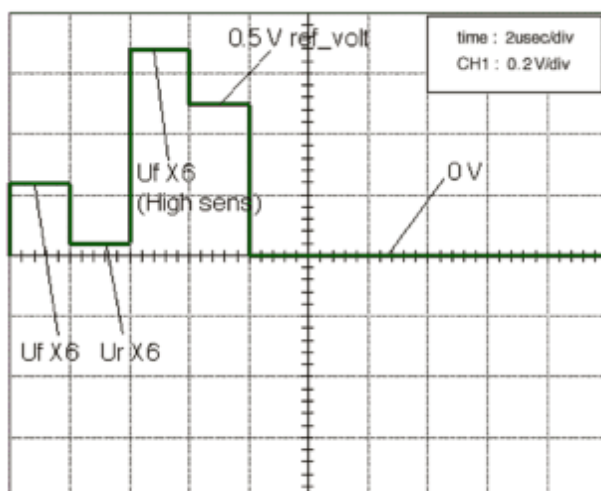
Sequence and meaning of the waveform segments

Signal from TALES	Duration (μ s)
Forward voltage X5-X6 (LC TX, high sense)	2
0.5V reference voltage	2
Forward voltage X1-X3 (BC 0-degree system)	6
Reflected voltage X3-X1 (BC 0-degree system)	2
Forward voltage X2-X4 (BC 90-degree system)	2
Reflected voltage X4-X2 (BC 90-degree system)	2
Forward voltage X5-X6 (LC TX)	2
Reflected voltage X6-X5 (LC TX)	2

MR system Symphony (TX to LC; transmitting to Extremity Coil)

- Set scope CH1 to 0.2 mV/div and time scale to 2ms/div.

Fig. 14: Transmitter amplitude 100V, transmitting to Extremity coil



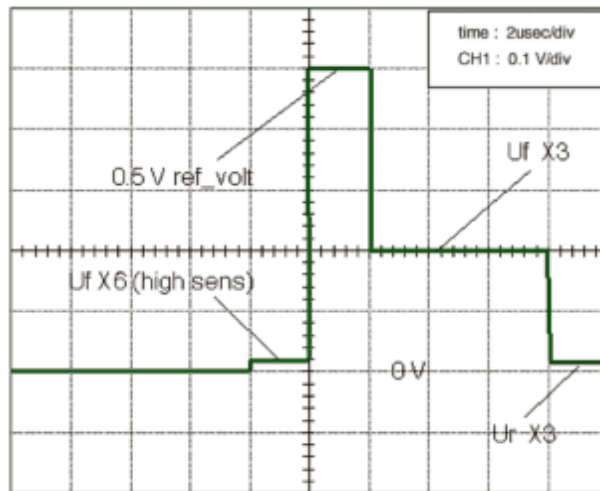
Sequence and meaning of the waveform segments

Signal from TALES	Duration (μs)
Forward voltage X5-X6 (LC TX)	2
Reflected voltage X5-X6 (LC TX)	2
Forward voltage X5-X6 (LC TX, high sense)	2
0.5V reference voltage	2
Forward voltage X1-X3 (BC 0-degree system)=0	6
Reflected voltage X3-X1 (BC 0-degree system)=0	2
Forward voltage X2-X4 (BC 90-degree system)=0	2
Reflected voltage X4-X2 (BC 90-degree system)=0	2

MR system Harmony (TX to BC; transmitting to BC)

- Set scope CH1 to 0.1 mV/div and time scale to 2ms/div.

Fig. 15:



Sequence and meaning of the waveform segments

Signal from TALES	Duration (μs)
Forward voltage X2-X4 (BC 90-degree system)=0	2
Reflected voltage X4-X2 (BC 90-degree system)=0	2
Forward voltage X5-X6 (LC TX)=0	2
Reflected voltage X6-X6 (LC TX)=0	2
Forward voltage X5-X6 (LC TX, high sense) -> 0	2
0.5V reference voltage	2
Forward voltage X1-X3 (BC TX)	6
Reflected voltage X3-X1 (BC TX)	2

2.7 RFCI/PIN-Booster (BC/LC)

2.7.1 PIN Diodes control

2.7.1.1 Prerequisite

- MR system is on.
- Start the Service Software and select "Test Tools".

2.7.1.2 Procedure

- Select RFCI
 - » Follow the instruction of the service program.

After the semi-automatic tests a test for manual DVM measurements is started:

- » Alternately two voltages will be present at the test points: - Reverse voltage of PIN diodes (UR) - Forward current voltage drop (IF) at PIN diode (according a forward current of 100 mA for the local coils. The service plug can be connected to the four plug positions successively without interrupting the test.

Check of the PIN Diodes of Body Coil Trio:

For a quick check of the PIN-diodes remove the connector X3 of the detuning unit at the BC and measure with the "diode test function" of a DVM in both (plus/minus) directions at the detuning unit.

Typical displayed values: forward direction: 0.411
 reverse direction: "OL"

Refer also to (→ Interactive Functional Description / MR-000.850.05).

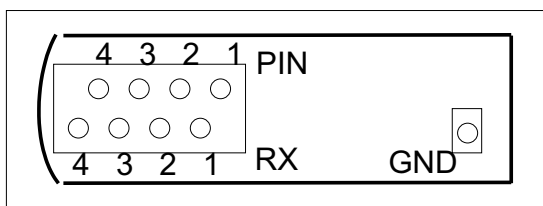
1. Measurements at the Service Plug:

The service plug is connected to coil connector (x=1...4):

Tolerance range:

- Reverse voltage (UR) at PIN diode: - 27 ... - 33 VDC
- Forward current voltage drop (IF): + 1.9 ... + 2.3 VDC

Fig. 16: Service plug - top view



- Measure the voltages at the four test points (PIN) of the service plug against GND.

- Measure the +10 (+/- 1) VDC RX-preamplifier supply voltage at the test points (RX) against GND. The voltage must be stable without drift and hum.
- For an exact measurement of the RX-preamplifier supply voltage, you have to abort the PIN diode test. (During the test, the voltage oscillates corresponding to the PIN diode switching).

2. Measurements at RFCI service board

Test points for local coil PIN diodes:

J10 left side TP 1-4 --> coil connector 1, PIN diode 1-4

J10 right side TP 5-8 --> coil connector 2, PIN diode 1-4

J9 left side TP 9-12 --> coil connector 3, PIN diode 1-4

J9 right side TP 13-16 --> coil connector 4, PIN diode 1-4

J7 left top TP --> body coil, PIN diode located in the BTB

Description:

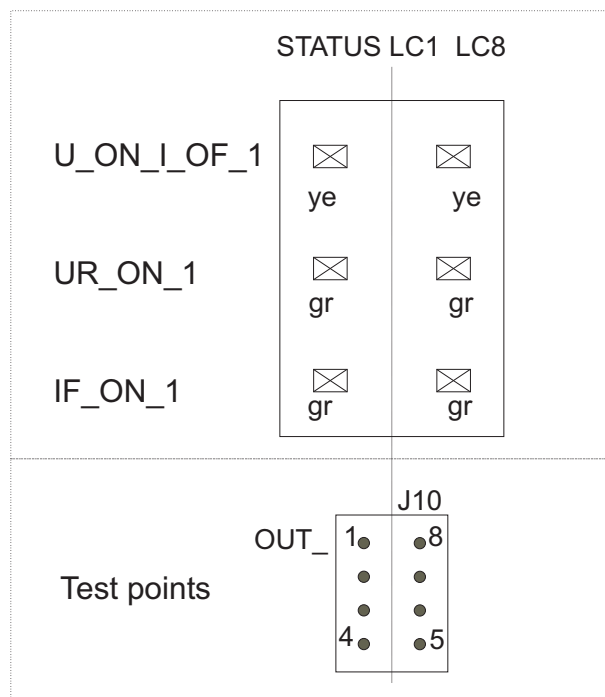
During the RFCI Interactive test the LED's at the RFCI service board alternating light up. There are 4x4 groups of LED's for each of the four PIN diodes connections of every coil plug. Fig. 9 shows the LED's for one PIN diode of two coil connectors.

U_ON_I_OF_1 on: PIN diode control active

UR_ON_x on: UR present at test point x against ground

IF_ON_x on: IF present at test point x against ground

Fig. 17: Service board RFCI: LED's of local coil PIN diode control and test points (partial view)

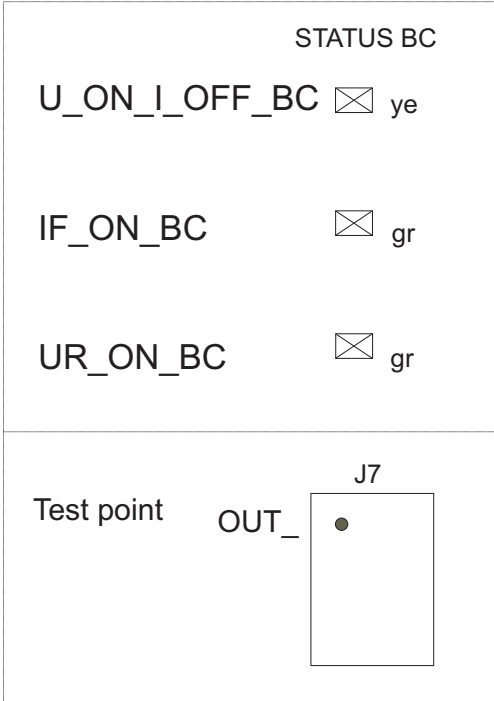


Tolerance range (local coil) PIN diode control:

- Reverse voltage at PIN diode (UR): - 27 ... - 33 VDC

- Forward current voltage drop (IF): + 2 ... + 6 VDC
(dependent on the cable length between CCA and patient table).
Error:
Open circuit voltage (IF current regulator at limit): approx. + 14 VDC
- Leave the voltmeter connected to a test point to see the two alternating voltages

Fig. 18: Service board RFCI: LED's of body coil PIN diode control and test points (partial view)



- Tolerance range (Body coil) PIN diode control:
- Reverse voltage (UR) at PIN diode: - 27 ... - 33 VDC
 - Forward current voltage drop (IF): + 3 ... + 8 VDC
(according to a forward current of 600 mA and dependent on the cable length between CCA and body tune box).
- Error:
Open circuit voltage (IF current regulator at limit): approx. + 15 VDC

Conclusions:

1. Measurement at the service plug:

UR	IF	Conclusion
OK	OK	OK
NOT OK	OK	Error RFCI
OK	NOT OK	Error RFCI; PIN diode of service plug may be also defective.

UR	IF	Conclusion
10 VDC (LC preamplifier)		Conclusion
NOT OK		1. Broken or shorted cable between LCCS (X11/12) and coil connector 2. 15V to 10VDC conversion at LCCS defective

2. Measurement at RFCI service board local coil:

UR	IF	Conclusion
NOT OK	OK	Error RFCI
OK	NOT OK	Broken cable between RFCI and LC-connector
NOT OK	NOT OK	Error RFCI or short circuit in the cable between RFCI and LC-connector

3. Measurement at RFCI service board (Body coil):

UR	IF	Conclusion
OK	OK	OK
NOT OK	OK	Error RFCI/ Pin diode in the Body coil defective (short circuit).
OK	NOT OK	Defective PIN diode of Body coil or broken cable between RFCI and Body tune box (BTB) or PIN-Booster (Trio) or between PIN-booster and Body coil. Check the LED's at the PIN-Booster (Trio). Perform TestTools/RF/BTB/BC/Functional.
NOT OK	NOT OK	Error RFCI or short circuit in the cable between RFCI and BTB or PIN-Booster (Trio), PIN-Booster and Body coil or defective PIN diode of body coil. Check the LED's at the PIN-Booster (Trio). Perform TestTools/RF/BTB/BC/Functional.

4. Additionally you can measure several supply voltages at the RFCI service board.

For further information about the **PIN-Booster** (meaning of LED's and test points) refer to (→ Interactive Functional Description / MR-000.850.05).

2.8 LCCS

- Check LED's on LCCS if RF Characteristic measurement does not work. (beside this, LED V91 on RFCI and LED S1 on TAS_C must be on during RF Characteristic measurement).

2.8.1 Meaning of LED's on LCCS

LED	ON	OFF
V87	+15V from RFCI to LCCS X36.1 (GND X36.2) OK	+15V from RFCI to LCCS X36.1 (GND X36.2) NOT OK
V88	-15V from RFCI to LCCS X36.3 (GND X36.2) OK	-15V from RFCI to LCCS X36.3 (GND X36.2) NOT OK LCCS no power, all LED's on LCCS must be off.
V89	+10V from LCCS X11/X12 to LC amp's; (center pin of coax connectors at +10V)	DC overload on one or more coax outputs LCCS X11/X12
V91	Relays S1 control voltage on; Status TAS_C, LCCS X35.4 (same status as S1/ TAS_C, Tx switched to dummy load TAS_C; service or rf characteristic mode)	Relays S1/LED off Tx switched to BC or LC (imaging mode)
V92	Relays S1 control voltage on; Status SAMI, LCCS X35.9 (only spectroscopy option) Open collector output on	Open collector output off
V93	Relays 2 control voltage on; Status TAS_C, LCCS X35.5 (same status as S1/ TAS_C, Tx switched to LC TX coil).	Relays S2/LED off Tx switched to BC.

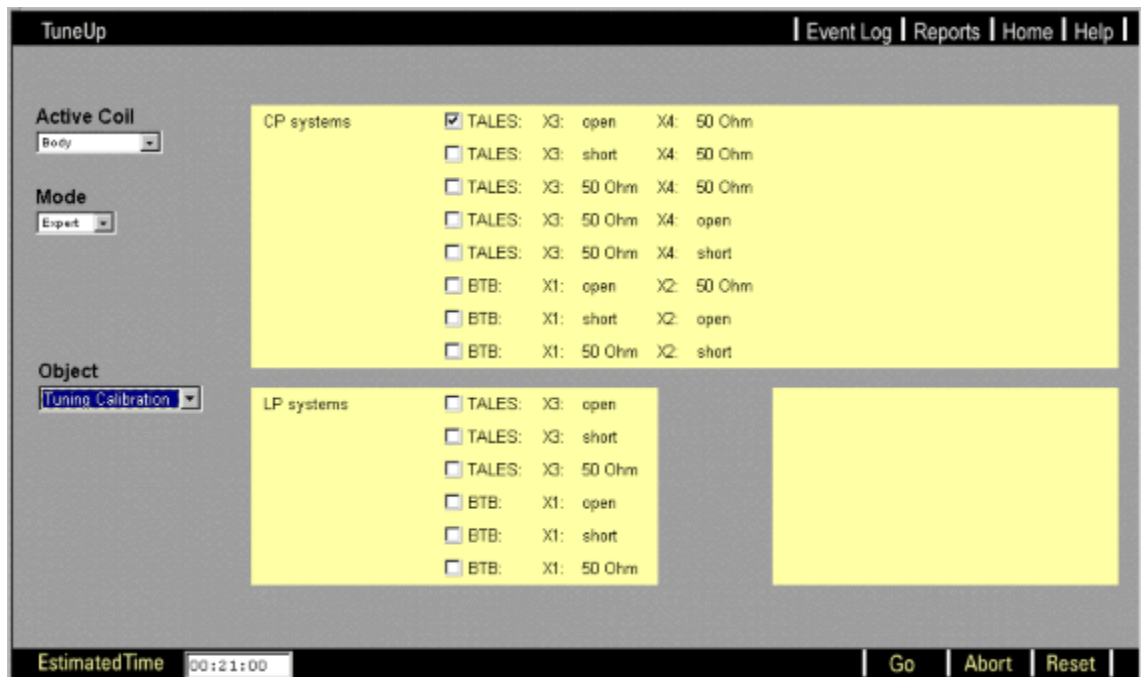
2.9 Additional Procedures

2.9.1 Tuning Calibration - Tune-up - Expert Mode

The tuning steps can be selected separately.

- » After successful completion of TALES calibration, the BTB and BC tests can be used. (If the BTB calibration is not done, the transmission evaluation (Symphony only) will fail).

Fig. 19: User interface Tuning calibration (Expert mode)



2.9.1.1 Common problems

- **External cable connections are not OK:**
 - **Check Transmit path:**
TX Module "TTX" -> CCA top "X60" -> RF-filter plate "X9" -> LCCS X30
 - **Check Receive path:**
RX Module "MR SIG 1" -> CCA top X51 -> RF-filter plate X1 -> LCCS X1
 - Check receiver channel signal input MR_Sig 2 (used on Symphony systems).
- **Bad 50 Ohm terminators:**
Use the FLUKE DVM to verify.
- **Wrong termination during the calibration:**
50 Ohms - short - open
- **Noise on CAN Bus caused by IOP board**
-> Check/replace IOP board.

2.9.1.2 Error Messages

If the error message indicates TALES.... or later on BTB..., the TALES or BTB itself may not necessarily be the problem. It can be any one of the items mentioned above. Certain tests are necessary to isolate the problem.

1. TALES-related Messages

"TALES Calibration data is not meaningful."

"TALES Verification is out of specification."

- » Check if the terminators used are OK.
Replace TALES if not successful.
- » Check cables

2. BTB-related Messages

If one or more "s-parameters" are out of spec, the following error messages will be generated during the "Tuning Calibration":

"S11_0 data is outside allowed tolerance"

"S22_0 data is outside allowed tolerance"

"DET_0 data is outside allowed tolerance"

"S22_0 data shows large deviation over frequency!"

For CP systems, additional error messages might be generated:

"S11_90..... tolerance"

"S22_90.....tolerance"

"DET_90.....tolerance"

"S22_90..... frequency"

- » Proceed as follows if any of these messages appears:
 - Run the RFPA/TALES tests.
 - Execute the "BC" test.

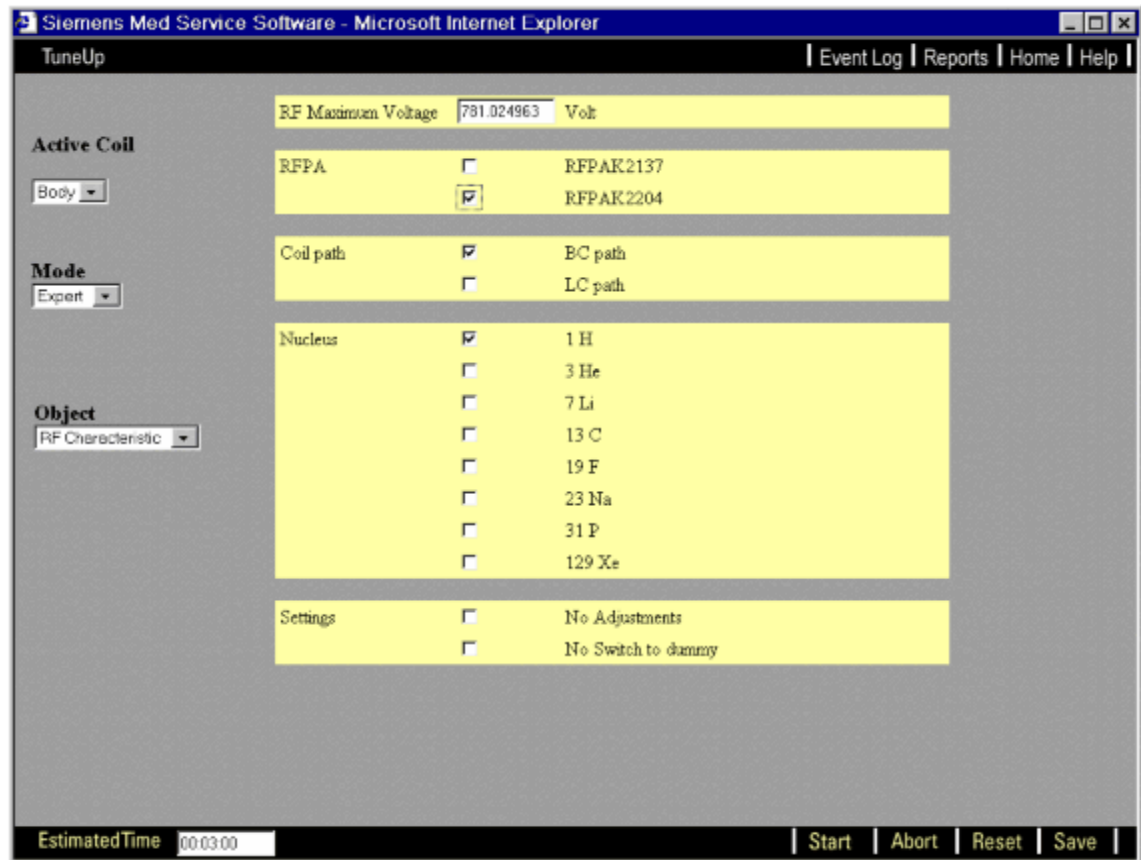
2.9.2 RF Chara - Tune-up - Expert Mode

Several measurement option can be selected:

- Maximum RF voltage
- Selection of RFPA (If spectroscopy option is installed)
- Selection of TX-path
- Selection of TX-frequency/nucleus (If spectroscopy option is installed)
- Measurement with no adjustments (frequency, ...)

- Disabling of automatic switching to dummy (TAS_C)

Fig. 20: User interface RF Characteristic (Expert mode)



2.9.2.1 Possible RF Chara measurements in case of problems

- Select a lower maximum voltage.
- Do the RF Characteristic measurement without adjustments (especially frequency adjustments) while using the TAS_C dummy load or an external 50 Ohm power dummy for the measurement.

(The frequency can be manually typed in: Options/Adjustments).

2.9.2.2 Error Messages

Frequency Adjustment:

“Frequency out of range”

The magnet current has to be adjusted (ramped up). Use the frequency from the shim result and calculate the new current setting:

$$J_{new} = J_{actual} \times \frac{f_{upperlimit}}{f_{actual}}$$

for HARMONY or 1 T

$f_{upperlimit} = 40.55 \text{ MHz}$

for SYMPHONY or 1.5 T $f_{\text{upper limit}} = 63.75 \text{ MHz}$
 without spectroscopy
 $f_{\text{upper limit}} = 63.70 \text{ MHz}$
 (with spectroscopy)
 ramp up at 63.45 MHz

for TRIO $f_{\text{upper limit}} = 123.450 \text{ MHz}$
 (without spectroscopy)
 ramp up at 122.95 MHz
 $f_{\text{upper limit}} = 123.265 \text{ MHz}$
 (with spectroscopy)
 ramp up at 123.10 MHz

(Tolerance range $f_{\text{upper limit}}$: +0/-5KHz)

“Signal too low”

- Check if the cabling between BCCS (X9/X3) and LCCS (X10/X9) is reversed.
- Phantom not centered: Reposition the phantom
- Gradient offsets: Switch off the gradient cabinet
- Field homogeneity out of spec: Reshim the magnet
- Receive path not OK: Use another coil for test purposes only
- Execute the BTB, BC and the LCCS/Service plug tests

Calibration Voltage Search:

“Measured voltage is less than min_voltage”

(min_voltage = 10V=

“Error in expected RF-output: Check RF-amplifier”

The actual voltage level at the TALES is less than 10V whereas the nominal start value is already 100V.

- Execute the RFchara tests (Testtools). Most likely, there is a problem with the TALES.

“Trying to set Voltage to TRA-Voltage”

“Cannot set to value greater than max_lim_voltage”

(which is 800V)

“Error in expected RF-output: check RF-Amplifier”

The system tries to find the nominal voltage, i.e. 566V for 1.5T- or 400V for 1Tsystems at the TALES.

This error occurs if the calculated /nominal value is > 800V in order to get the correct TALES value.

- Execute the RF chara tests. Most likely, there is a problem with the TALES.

“Number of allowed search measurements exceeded”

"Error in reaching or finding calibration voltage"

The search measurement makes 25 attempts to find the 566V/400V levels before the above error is generated.

There could be a RF instability problem either in the transmit or in the receive path.

- Execute the RF Characteristic tests (TestTools).

RF calibration:

If any of the following errors occur, it is necessary to check the RF power using a URV5. Most likely, there is a problem with the RFPA DICO.

- Execute the RF Characteristic tests.

"Value of PF ADC Conversion is too high" >= 0.2**"Please check the RF-components; especially the RFPA DICO"**

The DICO factor will be calculated using the "voltage search" results:

$$\text{DICO - Factor} = \frac{\text{TALES Voltage}}{\text{DICO - ADC}}; \text{ it must be } < 0.2$$

"ADC value of ADC is greater than maximum"**"Calibration is too high: Please check RF-components; especially RFPA DICO")**

The expected ADC value is out of range; ADC overflow.

$$\text{ADC}_{\text{max}} = \frac{\sqrt{\text{MaxPower} \times 50 \text{ Ohm}}}{\text{DICO Factor}}$$

"Value of peak power is too high"**"Please check RF-components"**

There is a difference between the nominal value and the value measured with the DICO. Most likely, there is a problem with the DICO (stability or linearity).

RF Max Power search:

"Measured voltage is less than "min.Voltage"; min.Voltage = 10 V

"Error in expected RF-output; check RF-Amplifier"

One of the involved components failed just now; most likely the RFPA.

- Execute the "RF-Chara"-test routines.

"Trying to set voltage to "TRA-Voltage"

"Cannot set to value greater than max. linear voltage"

"Error in expected RF-output! Check RF-components"

The RFPA does not reach specifications. The reason could be:

The level of the SSB-signal (TX-Module) or the RFPA/DICO itself is faulty.

- Execute the "RF-Chara"-test routines.

"Number of allowed search measurements exceeded"

"Error in reaching or finding maximum power"

"Please check RF-Amplifier and RF-Components"

With the previous measurement it was possible to reach the max. power. There could be an instability-problem in the transmitter or even in the receiver path.

- Execute the "RF-Chara"-test routines.

2.9.3 PerCom

This tool allows direct control of components, e.g. switching the TX-path (BC/LC/Dummy). Select the component and action typing in the appropriate number in the PerCom menus.

Fig. 21: PerCom start menu

```

MS-DOS Command Prompt - percom
Microsoft(R) Windows NT(TM)
(C) Copyright 1985-1996 Microsoft Corp.

C:\>percom
Enter password: *****
Warning: Class PCILinkProxyTest does not exist in RuntimeLibrary!
Warning: Class GCUProxyTest does not exist in RuntimeLibrary!
Warning: Class TXUProxyTest does not exist in RuntimeLibrary!
Warning: Class RXUProxyTest does not exist in RuntimeLibrary!
Warning: Class EDDYProxyTest does not exist in RuntimeLibrary!
Warning: Class BTBMPProxyTest does not exist in RuntimeLibrary!
Warning: Class PALIProxyTest does not exist in RuntimeLibrary!
Warning: Class DBUProxyTest does not exist in RuntimeLibrary!
Warning: Class STIMOProxyTest does not exist in RuntimeLibrary!

-----
Periphery Proxy Classes
-----
 1. RFCI
 2. IOP
 3. RFPa
 4. MPCU
 5. SCT
 6. SLIO
 7. DOOR
 8. TRa1
 9. CAN
10. Supervision
11. AlarmBox
12. RCA
13. MSUK2200
14. BTB1
15. BTB2
16. PARIO
17. RFASK2200
18. PTABK2200
19. CABUIK2207
20. GPAK2217
21. SHIM
22. PMU

-----
Please enter number or unique text of a cmd. <'q' for quit>
PERCOM >>_

```

2.10 Flowcharts for special problems

2.10.1 "Excessive reflected RF pulse power was detected by TALES"

This message appears within System/Report and within SeSo -> EventLog.

It is very likely caused by arcing components (e.g., bad cable connection/contact) in the transmit path (RF cables between RFPA and body coil).



To disable adjustments and/or safety supervisions (such as RFSWD or GSWD) during the following measurements refer to (→ Tools / Page 77) and following.

Fig. 22:

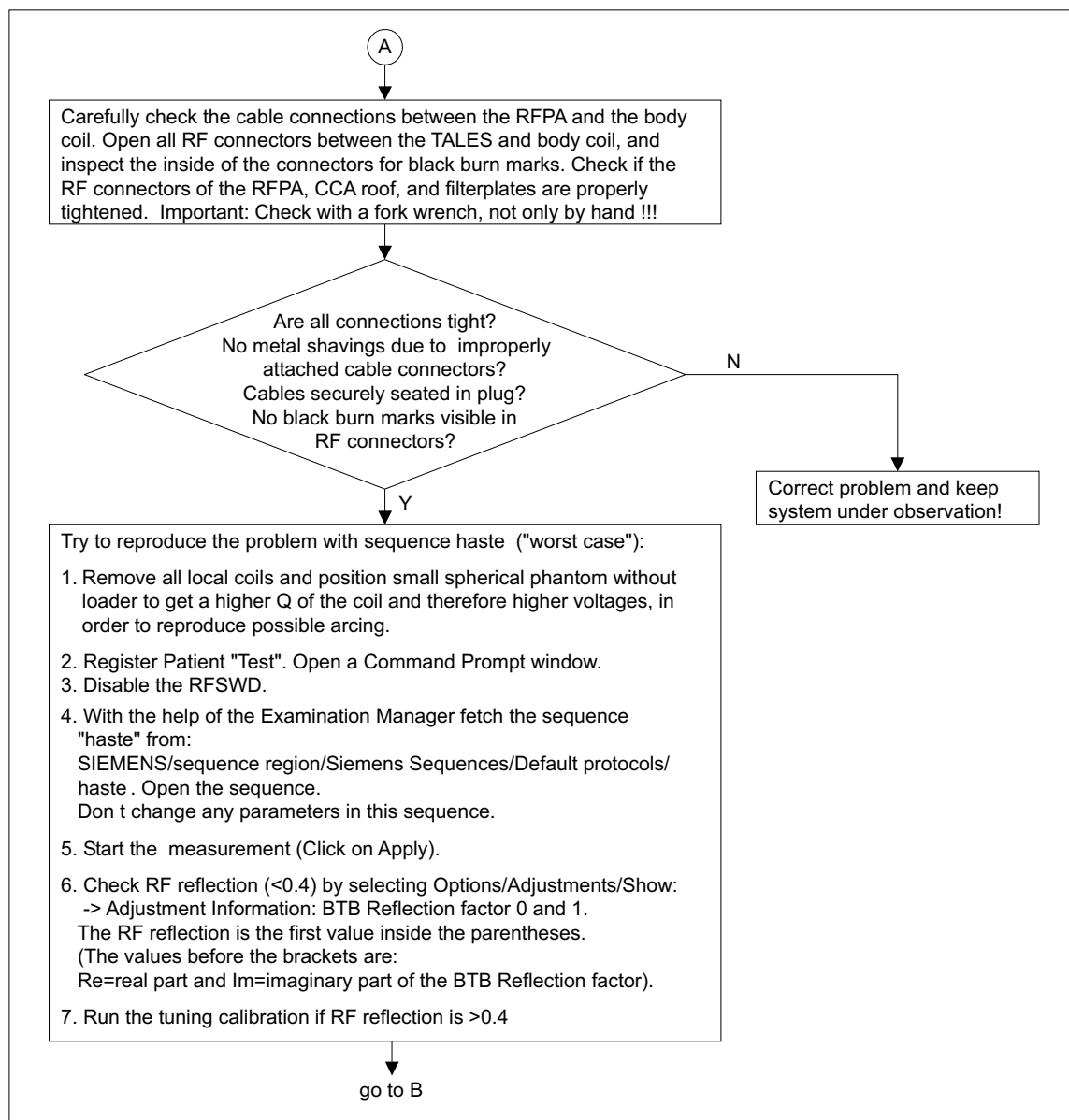


Fig. 23:

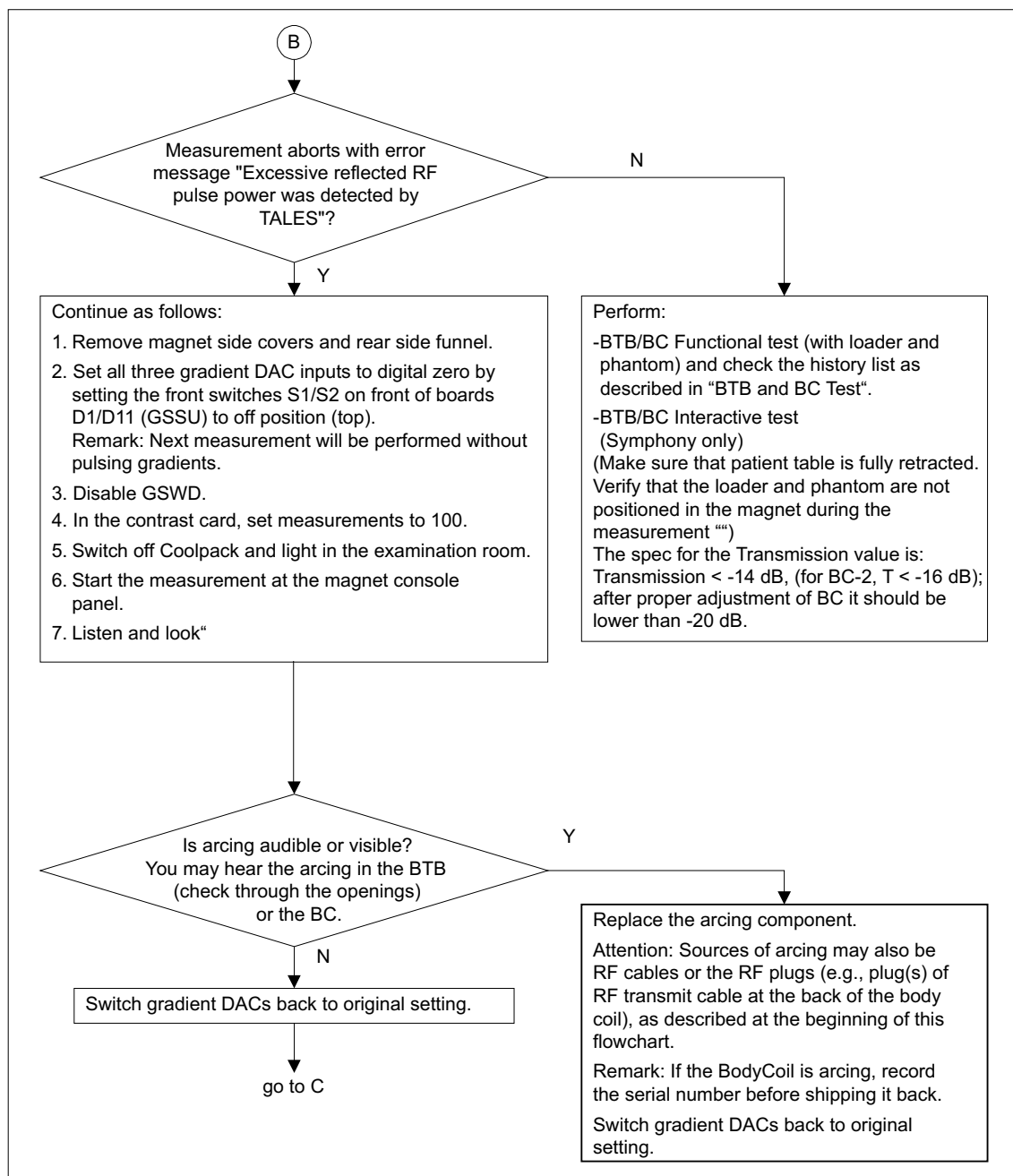


Fig. 24:

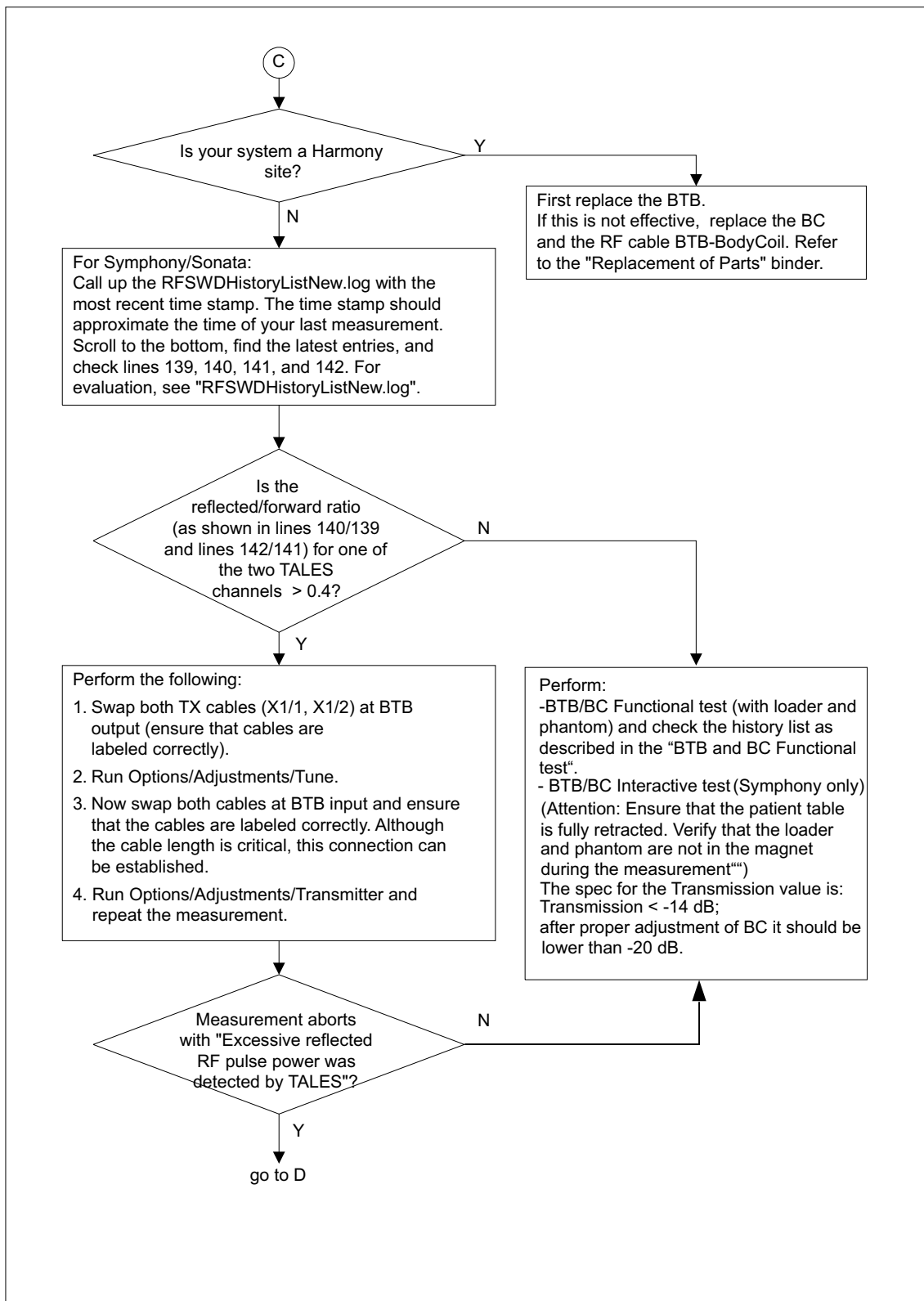
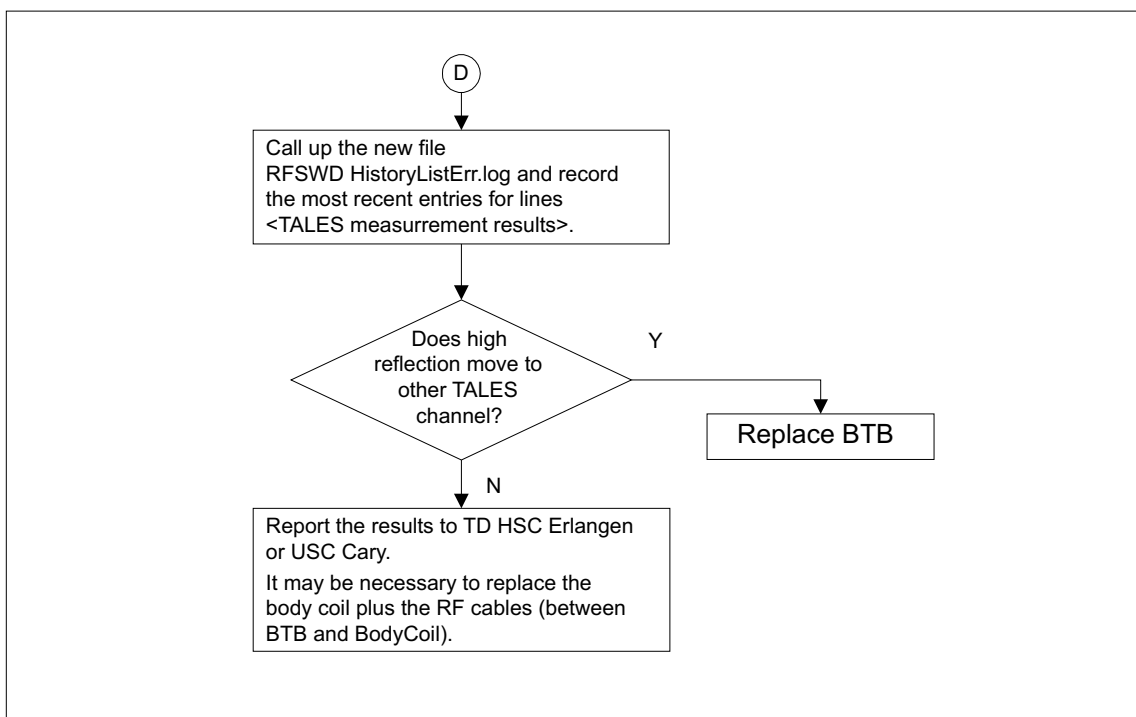


Fig. 25:



2.10.2 Check of 50 Ohm/Short Service Adapters (failed Tuning Calibration)

NOTICE

Use of defective 50 Ohm/Short service adapters leads to invalid Tuning Calibration results.

Failing or invalid Tune-up step Tuning Calibration or BC Tuning. Reduced TX-performance of the system.

- » For Tim-systems, the Tune-up step Tuning Calibration will fail completely when defective RF-service adapters are used.
- » Nevertheless, with minor deficiencies, Tuning Calibration can pass but provides wrong calibration data or leads to a failing BC Tuning.
- » Before use, always check condition and measure resistance of all service adapters/cables with a DVM.
- » In case of any deficiencies, order/use new service adapters.

2.10.2.1 Problem Description

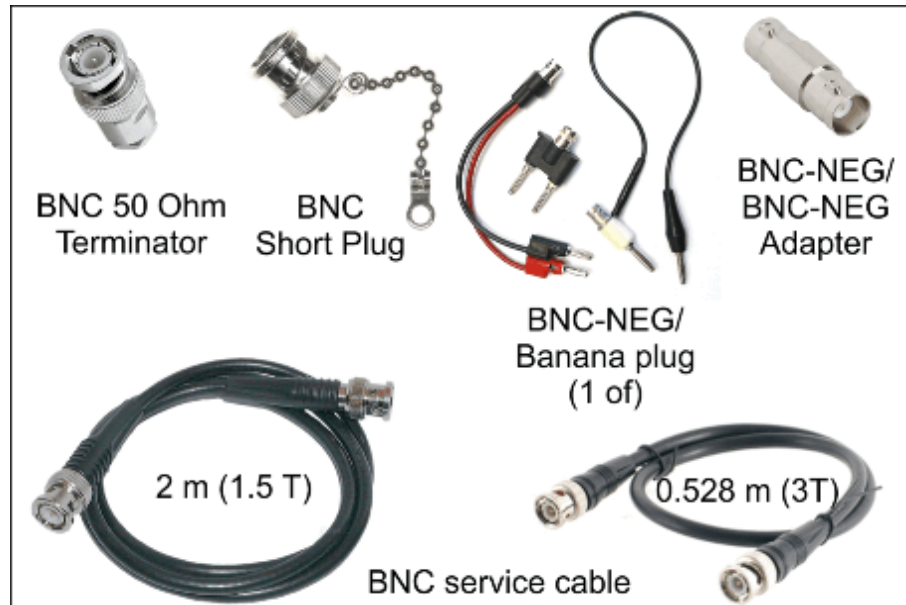
Use of defective RF service adapters can lead to multiple problems which are difficult to troubleshoot, e.g.:

- Failing Tune-up step Tuning Calibration (due to e.g. wrong resistance value of 50 Ohm terminator, broken BNC service cable, etc.).

- Unnecessary spare part consumption (e.g. new TALEs sent back as “dead on arrival” due to failing Tuning Calibration).
- Wrong display of BC Tuning values due to bad Tuning Calibration (Tuning Calibration is the reference for BC Tuning).
- Reduced TX-performance of the system due to wrong BC Tuning Adjustment as a result of a bad Tuning Calibration.

2.10.2.2 Required Tools

Fig. 26: Required Tools



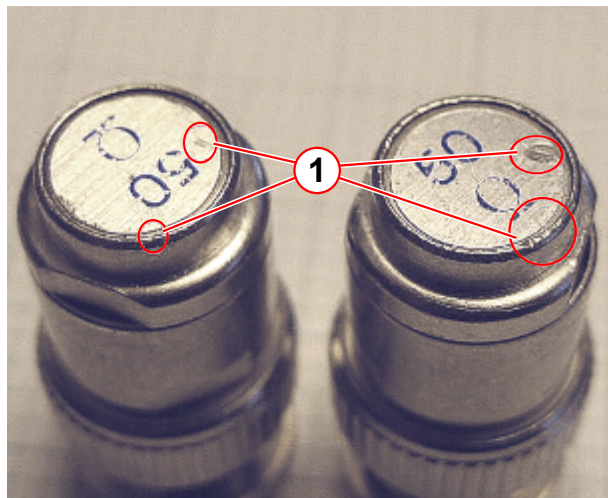
- Digital Multimeter FLUKE 287 or equivalent with a valid calibration
- Adapter BNC-NEG/Banana plug (30 25 749), part of service tool kit
- Adapter BNC-NEG/BNC-NEG (77 60 481), part of service tool kit
- 2 x BNC 50 Ohm terminator (31 10 392)
- 2 x BNC Short plug (31 63 714)
- BNC service cable used for Tune-up step Tuning Calibration:
 - BNC 2 m cable for 1.5T systems (87 33 479)
 - BNC 0.528 m cable for 3T systems (109 15 732)

2.10.2.3 Visual Checks

- Check condition of cables/adapters visually:
 - » There must be NO damage or excessive wear.
 - » Only use adapters which are in a good condition.
 - » Below, there are some examples for connectors which are NOT ok:

Examples of defective adapters → replace:

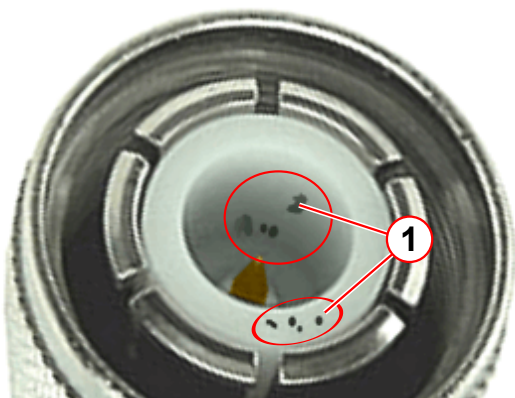
Fig. 27: 50 Ohm BNC Terminators



(1) Notches

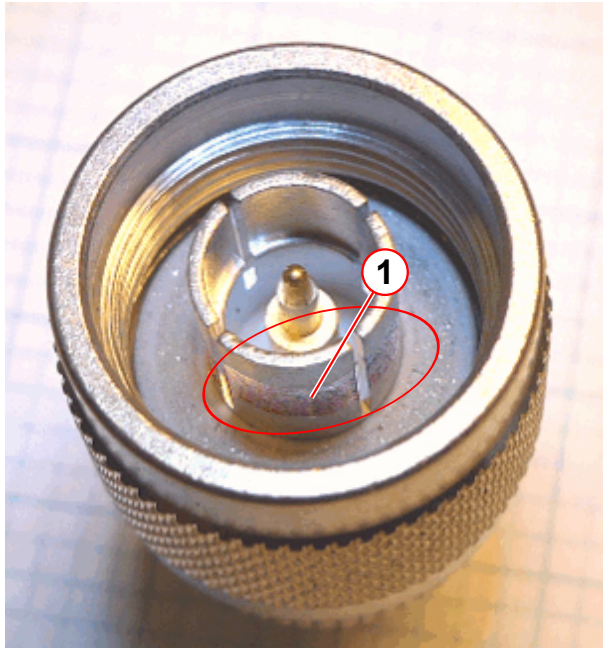
- Signs of mechanical damage, e.g. notches at dropped adapters:
 - » Handle adapters carefully.
 - » Store and transport adapters in a suitable box (e.g. with foam protection).

Fig. 28: BNC-HN Adapter

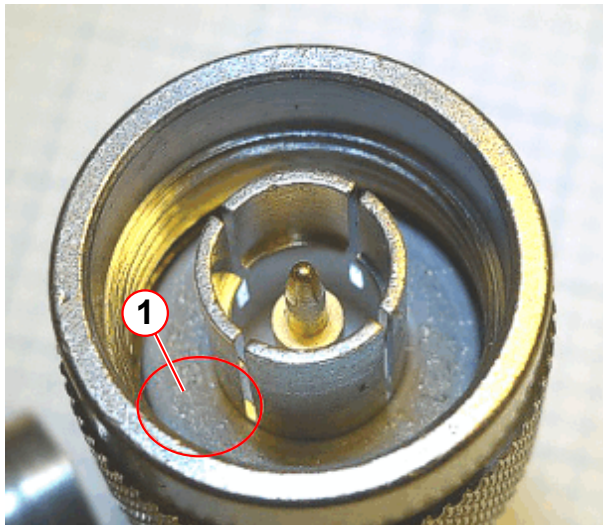


(1) Dirt

- Contamination with dirt, liquids, etc.:
 - » Work in a clean environment.
 - » Store, transport adapters in a protective packing.

Examples of defective adapters -> replace:*Fig. 29: BNC Adapter*

(1) Scratches outer conductor

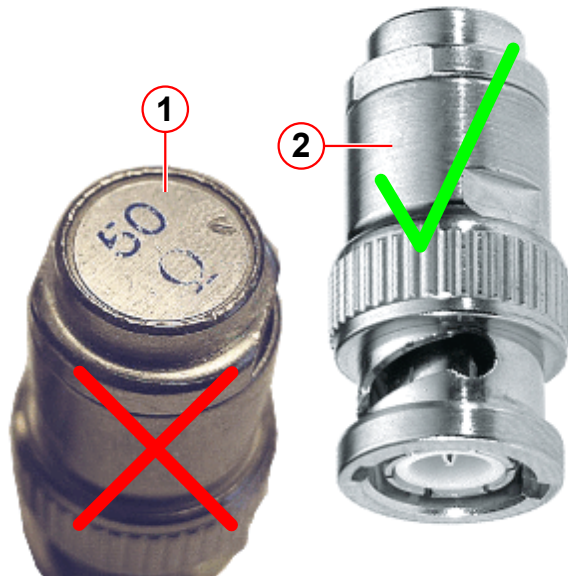
Fig. 30: BNC Adapter

(1) Metal dust

- Connector metal abrasion, e.g. scratches at outer conductor:
 - » Hold adapters when tightening - avoid turning.

Examples of defective adapters -> replace:

Fig. 31: 50 Ohm BNC Terminator



- (1) Oxidized - do NOT use!
(2) Shiny - good condition

- Oxidization:
 - » Scrap old, oxidized adapters - only use new ones.

2.10.2.4 Measurement Procedure with DVM

Step 1:

Fig. 32: Resistance measurement 50 Ohm BNC terminator



Measure resistance of **BNC 50 Ohm** terminators:

- Use DVM with adapter BNC-NEG/Banana plug.
 - » The resistance must be 50 ± 0.5 Ohm.
 - » Otherwise, replace adapter before use.

Step 2:

Fig. 33: Resistance measurement BNC Short Plug



Measure resistance of **BNC Short Plug**:

- Use DVM with adapter BNC-NEG/Banana plug.
 - » The resistance must be $0 + 0.5 / - 0$ Ohm.
 - » Otherwise, replace Short Plug before use.

Step 3:

Fig. 34: Resistance measurement BNC Service cable

Measure resistance of **BNC Service Cable**:

- Use DVM with adapter BNC-NEG/Banana plug.
- Connect service cable to DVM adapter.
- Connect adapter BNC-NEG/BNC-NEG to the cable end.
- Connect BNC Short Plug.
 - » The resistance must be $0 + 0.5 - 0 \text{ Ohm}$.
 - » Otherwise, replace BNC Service Cable before use.

2.10.3 "RFPA Power Fault" / "RFPA Overcurrent"

(CANSLIO_RFPARFPOWERFAIL **ID 27**, Power failure message from RFPA/ CANSLIO_RFPA-CURRENT **ID 30**, RFPA power supply current too high/ CANSLIO_RFPARFPOWER **ID 31**, power too high)

Fig. 35:

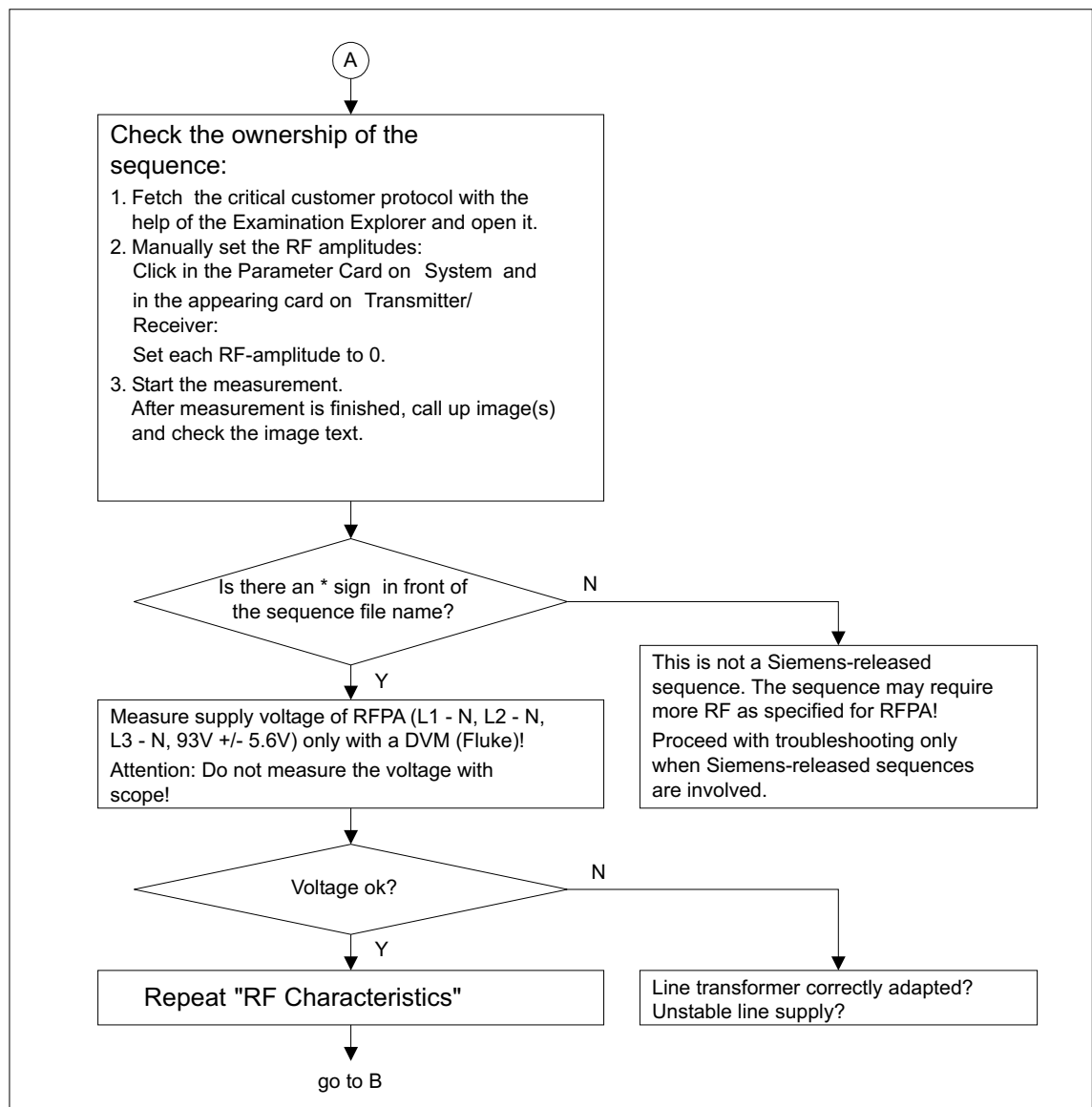
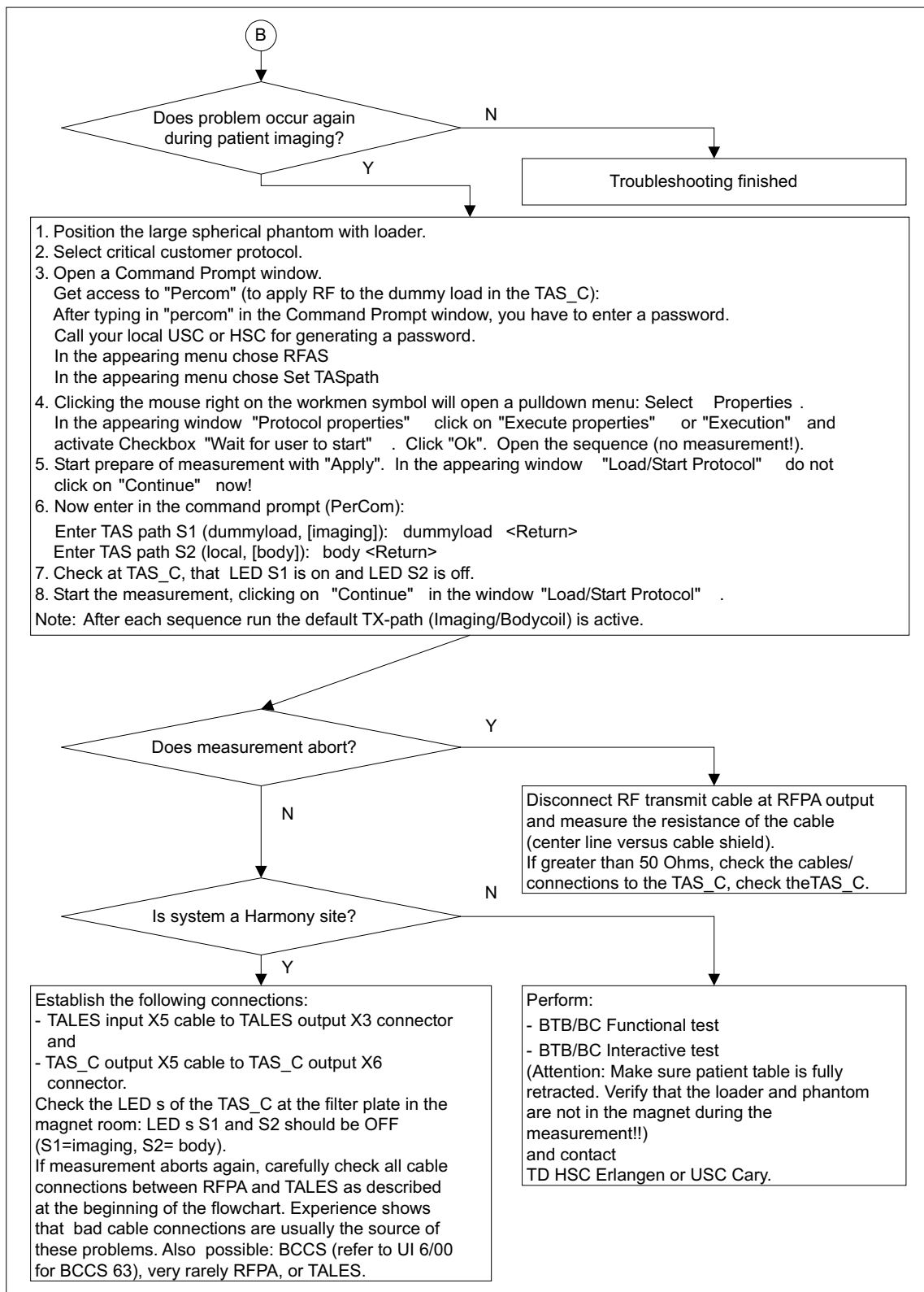
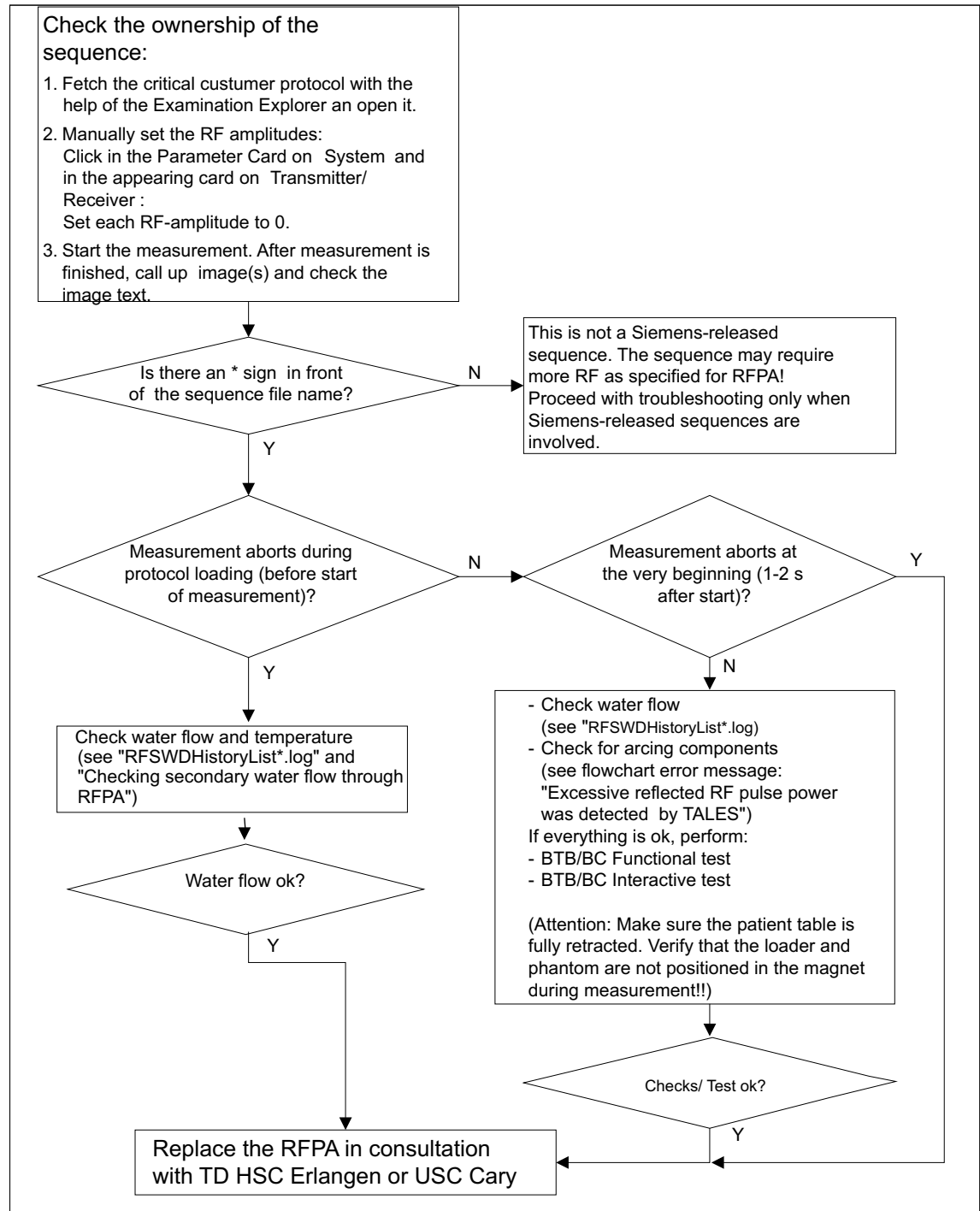


Fig. 36:



2.10.4 "RFPA Overtemperature"

Fig. 37:



2.10.4.1 RFSWDHistoryList*.log

This file can be called up with SeSo -> "Reports," menu item "Process diagnostics."

Whenever the patient is changed (Pat/Reg) and the first protocol is started, a new file RFSWDHistoryListNew.log is generated and the original file is copied to RFSWDHistoryListOld.log.

In the case of high RF reflection, for example, the file is also saved as RFSWDHistoryListErr.log. Select one of these files and scroll down to the bottom (the file may be quite large). In line 139 and 140, the peak maximum forward and peak maximum reflected power measured by TALEs channel 1 are shown in Watt. In line 141 and 142, the same values measured by TALEs channel 2 are shown.

Example (ok condition!, the line numbers are system-dependent):

<TALES Measurement Results>

153 TALESPpeakMaxForw_1.....: 1338.660400 [W]

154 TALESPpeakMaxRefl_1..... : 84.671265 [W]

155 TALESPpeakMaxForw_2.....: 1287.063843 [W]

156 TALESPpeakMaxRefl_2..... : 218.698105 [W]



The absolute power values depend on the selected sequence. The forward/reflected ratio of each channel indicates the quality of impedance matching.

2.10.5 Checking secondary Water Flow through RFPA

Open the water connection at the RFPA (DORA) output and measure how many liters flow into a bucket in 1/2 min. Given the nominal value of 8 l/min, it should be at least 4 liters.



Two quick-release water couplings are needed to conduct this test: Material number 5940726 (male coupling) and material number 5940718 (female coupling).

If the water flow is not ok and your system is a Symphony Quantum, refer to UI 2/00. For Symphony and Sonata systems, refer to UI 11/00.

2.11 New RF Test Procedures (TestTools)

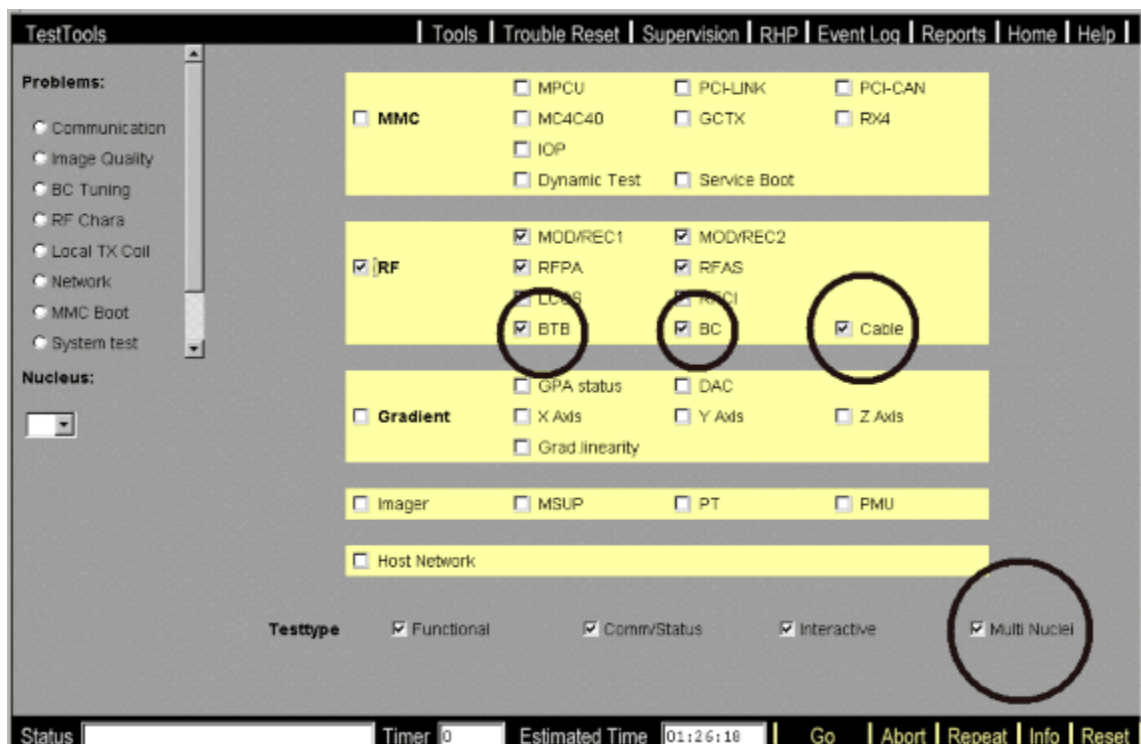
2.11.1 General

This section describes additional tests for troubleshooting on components (FRU's) of the RF-System, such as components in the TX-path (RFPA, TALES, BTB, BC and RF-cables).

The following test are available beginning with software version VA21A (syngo MR 2002A) in the Service Software under TestTools:

- Tools (switch off supervisions, additional options)
- RF/BTB
- RF/BC
- RF/cable
- Multinuclei Tests

Fig. 38: New RF-tests under TestTools



2.11.2 Procedures

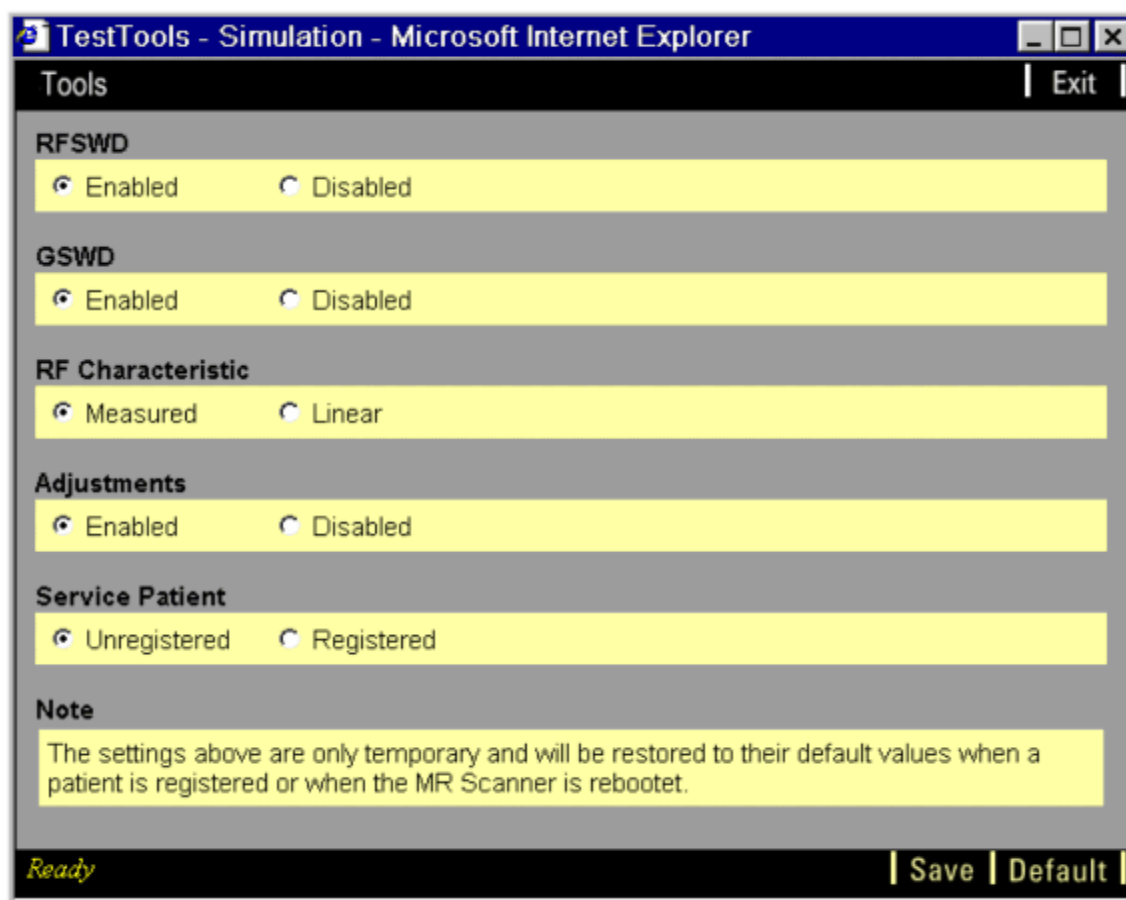
2.11.2.1 Tools

The user interface offers the following options:

- RFSWD disabled: switch off power limitation when using high-power measurements.
- GSWD disabled: Switch off gradient strength limitation by the stimulation monitor
- Selecting of measured or linear RF characteristic

- Switch off of adjustments preceding a measurement
- Registering/de registering of the "service patient"

Fig. 39: Tools user interface window



2.11.3 BTB and BC Tests

2.11.3.1 Prerequisites

- Valid tuning calibration; TALES calibration is sufficient
- TestTools test must be ok: RX4, MOD/REC, BCCS/LCCS loops
- The BTB and BC tests (except the BC power test) use low voltage RF-signals using the TTX1 output of the TX-module via LCCS, BCCS back to the RX-module. The BTB tests are performed at the system center frequency (40.45 MHz resp. 63.6 MHz).

2.11.3.2 Test Procedure

The basis for the tests is the measurement of reflection factors, which are complex numbers.

During the tests the setting of the serial and parallel capacitors and even detune settings are changed.



For functional tests ensure that the loader with the large spherical phantom is precisely positioned in the body coil.

2.11.3.3 BTB

■ BTB Tests functional

The BTB tests are based on the analysis of reflective voltage factors for the different switching modes of the short circuit switch, the PIN diodes and the tuning capacitors in the BTB.

The test consists of several individual tests each including a reference measurement and the actual level measurements.

Signal Path for the reference measurement

The TTX output signal of the modulator is fed to the LCCS-BCCS (preamplifier)-LCCS and to the receiver.

The level measured is used as the reference.

Signal Path for the measurement

The TTX output signal of the modulator is fed to the LCCS-BCCS-TALES-BTB-BC and reflected back to the BTB-TALES-BCCS-LCCS and receiver. The short circuit switch, the PIN diodes and the capacitors in the BTB are switched in sequence.

Tests:

1. Reflection factor measurement
2. Test 1-4: Static detuning active, the serial capacitors are switched in sequence.
3. Test 5-8: Dynamic detuning active, the serial capacitors are switched in sequence.
4. Test 9-13: The parallel capacitors are switched in sequence.

Conclusions:

The reflection factors (real and imaginary) are generated for the individual switch settings and checked against the specifications.

Afterwards, several relative values from the individual results are determined. They allow a more detailed evaluation of the BTB functionality.

The results of measurements 9-13 are displayed graphically in a x-y diagram (real part on x-axis, imaginary part on y-axis). From deviations of the measurement compared with the normal situation shown in the error can be analyzed (→ Fig. 41 Page 80)

Fig. 40: Evaluation BTB test 9-13

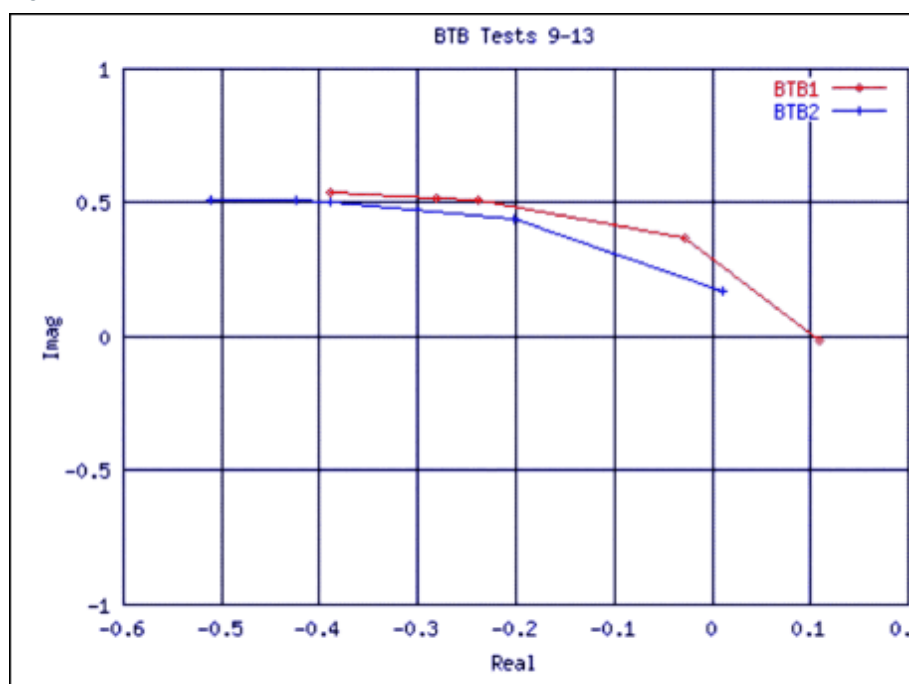
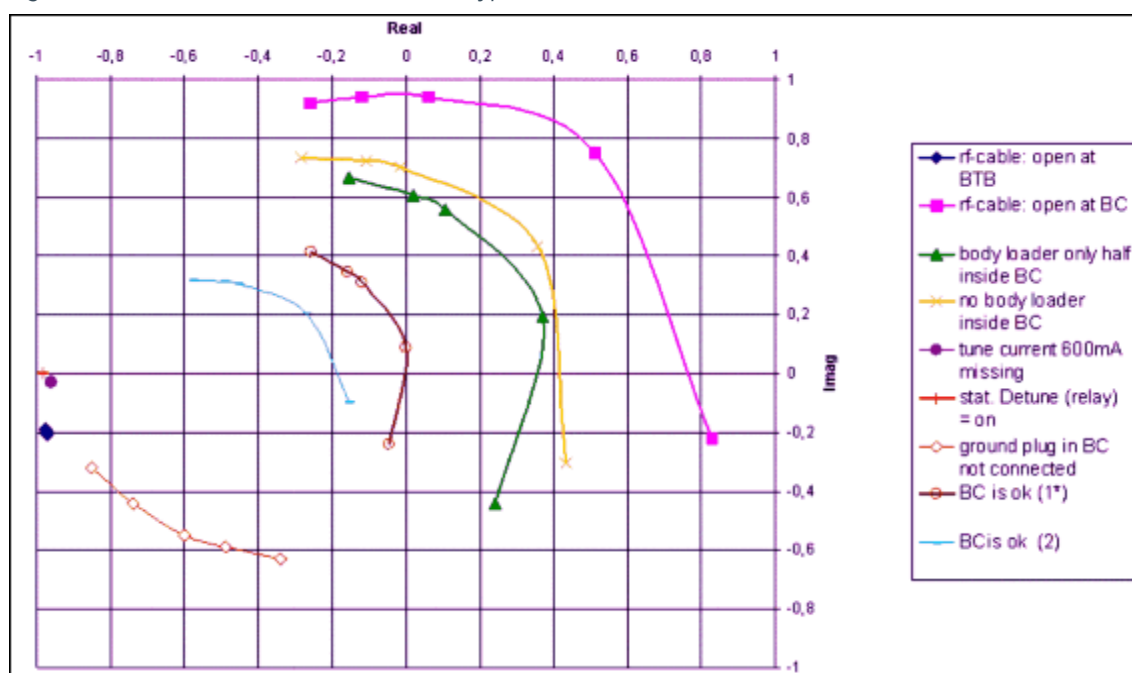


Fig. 41: Measurement results test 9-13 for typical BC errors



BTB Test interactive

1. The measurement of the BTB reflection factors is identical to the functional test but will be done without phantom and loader.
2. The measurement of reflection factors is identical to the functional tests but is done without phantom and loader.
3. The first test checks the "Tuning calibration" with TALES outputs X3 and X4 open.

4. The second tests with BTB outputs X1/1 and X1/2 outputs shorted checks the parallel capacitors C_s and relays of the BTB by measurement of the reflection factors of the BTB.

Conclusions:

- Check the correct BTB type in the configuration window.
- Repeat the Tuning calibration.
- If the 2nd test fails the BTB may be defective.

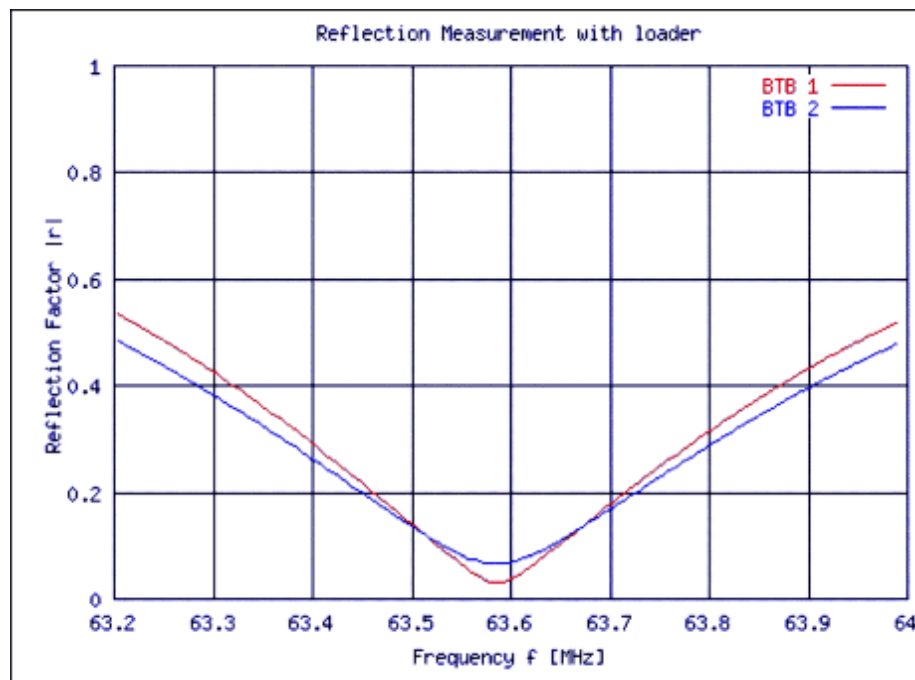
2.11.3.4 BC

■ Body coil Functional

(with large spherical phantom and loader)

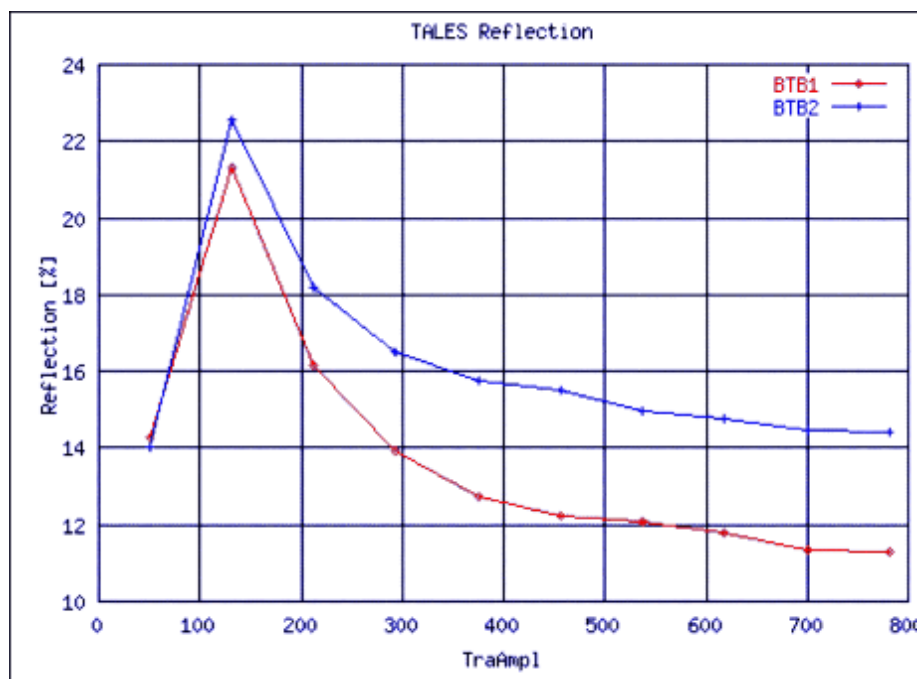
The body coil test is based on the analysis of reflective voltage factors that depend on the frequency. The test simulates a network analyzer that provides the frequency for the smallest reflection factor and the resonance curve of the body coil as results. The available frequency range of approximately 1MHz is stepped through in 10kHz steps. The respective reflection factors are produced and entered in a frequency reflection factor diagram. The frequency with the smallest reflection factor is checked against the specifications. The Q-factor of the coil will be evaluated and checked against specification.

Fig. 42: Reflection factor measurement (Example Symphony)



Additionally a power test with increasing transmitter amplitude is performed. The TALEs reflection values over the transmitter amplitudes are graphically displayed.

Fig. 43:

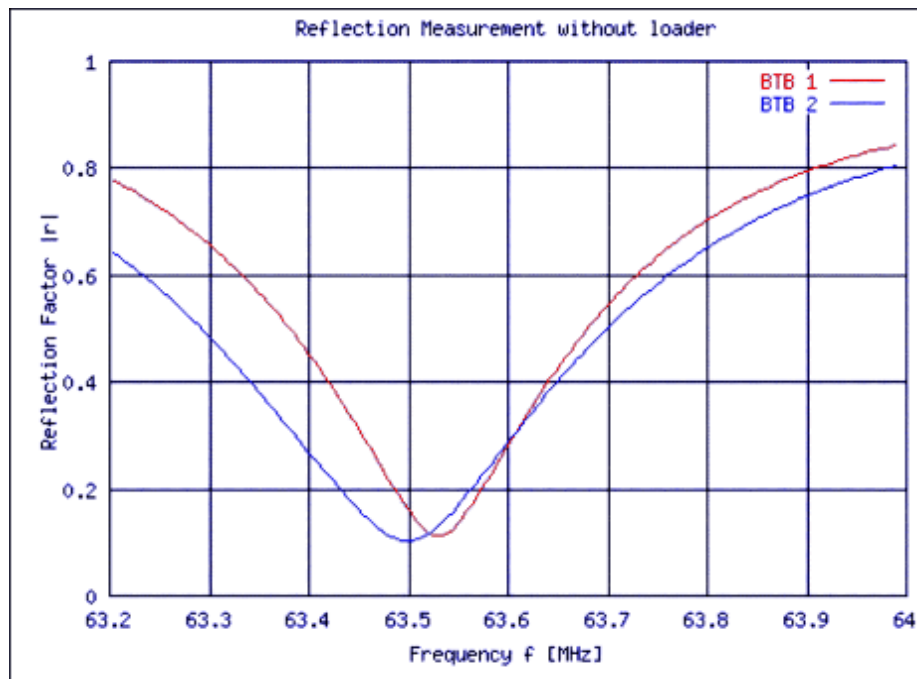


■ Body coil interactive

(without phantom and load)

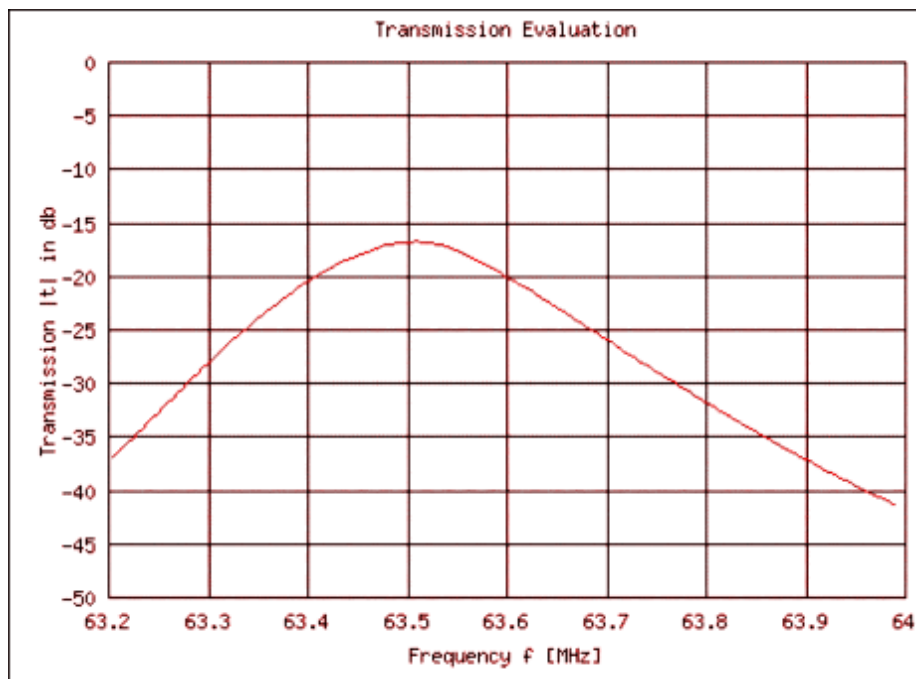
The first part of the measurement is identically to the functional measurement.

Fig. 44:



Additionally the decoupling of the body coil elements (transmission measurement only for Symphony) is evaluated.

Fig. 45:

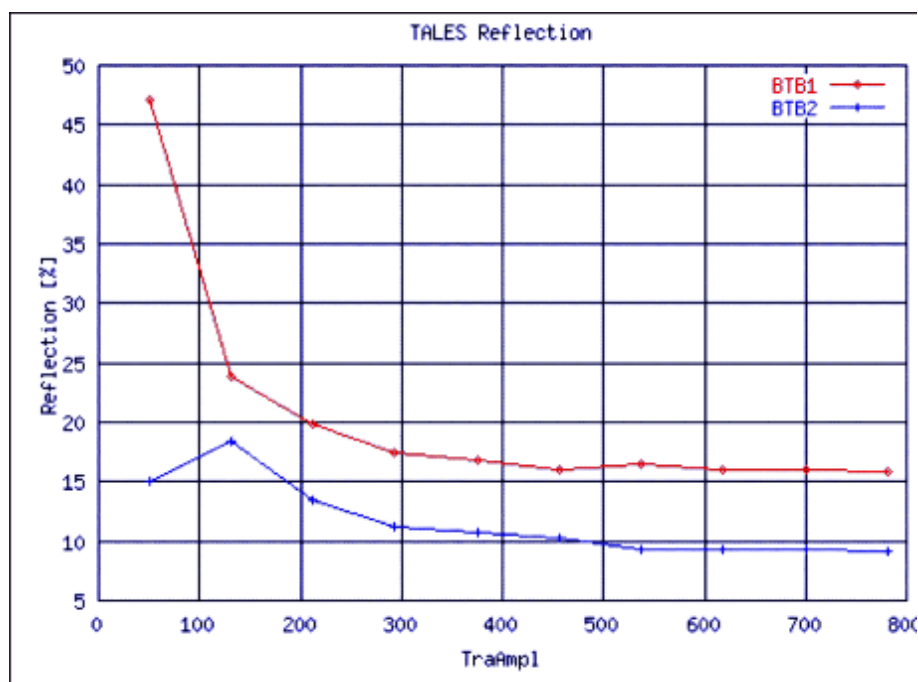




For body coil type 2 adjustments, refer also to (→ RF System / MR-000.841.14) Replacement of the Body Coil and Tune-up procedure "BCTuning".

Like the functional test a power test with increasing transmitter amplitude is performed. The TALEs reflection values over the transmitter amplitudes are graphically displayed

Fig. 46:



Conclusions:

If the tolerances of the reflection measurement evaluation are exceeded and the BTB tests are in specification, the problem is probably located in the BC.

The power test checks the disruptive strength of the BC. The test should run without shut-down to the maximum voltage. The reflection of TALEs should be approximately constant at voltages higher than 200 V (refer to (→ Fig. 46 Page 84).

The power test without load is the worst case, i.e. the reflected voltages are significantly higher.

Cable

- BC-TX-path (refl.)
- BC-TX-path (volt.)
- LC-TX-path (refer also to the Online Help)



These tests are for experts only!

Additionally special equipment is necessary for the tests. Special tools (power dummies, rf-cables, connectors, adapters and attenuators) are needed.



Whenever a dummy load is used to interrupt the TX-path BEFORE the TALES, the comparison of DICO- and TALES-values will not be successful and the test will fail.

This is normal and not a problem of the system

The tests enable you to check the complete TX-path to BC or LC and to make a systematic troubleshooting in the TX-path by connecting a power 50 Ohm terminator at various locations in the TX-path and/or configuration of particular loops.

The test should be used by experienced service engineers. Special tools (power dummies, RF-cables, connectors, adapters and attenuators) are needed.

Popup windows inform you about the actual measurement configuration.

Additional required Tools

- RF Adapter set (73 85 896)

Safety



WARNING

High RF voltage!

Contact with RF conductors during transmitting of the RFPA will result in death or serious injury.

- » Before disconnecting RF transmit cables or connectors in the transmit path, disconnect the RF input and the unblank line U1 from the RFPA.

NOTICE

Never use the circuit breaker F8 in the CCA cabinet to switch the RFPA ON or OFF if the MR system is powered ON.

Switching the RFPA ON or OFF via circuit breaker F8 in the CCA cabinet could damage the RFPA.

- » Switch the MR system to "standby mode" or completely OFF.

NOTICE

Be careful not to overload the signal inputs when establishing non-standard measurement loops!

Choose the transmitter voltage, repetition time and pulse time so that the dummy load and attenuators are not overloaded.

- » Always start with low transmitter amplitudes (approx. 50V). The pulse amplitudes must be checked/set for each new measurement.

Examples of breakpoints in the TX path

Location of dummy termination in the TX path	Symptom	Possible error	Service action
RFPA out	Power output too low	RFPA, DICO	Replace RFPA
CCA top out	High reflection DICO	RF cable CCA	Replace RF cable
TAS_C out (X4) TAS_C out (X5)	No signal/ low power	TAS_C control TAS_C switch	Check TAS_C power, TAS_C control (LED's). For path switching use PerCom. Check resistance of switch path with an Ohmmeter. Replace defective TAS_C
BCCS in (X1)	High reflection DICO	RF cable/ connectors	Replace RF cable/ connectors
BCCS out (X2, X8)	Low power High reflection DICO	Defective T/R switch	Replace BCCS

Location of dummy termination in the TX path	Symptom	Possible error	Service action
TALES out (X3, X4) TALES out (X6)	High reflection TALES DICO OK	Oscillation of RFPA/RFPA trans- mits not only MR frequency	Check RFPA, TAS_C, BCCS, cables, TALES
	High reflection TALES and DICO	Cable connec- tions TAS_C to BCCS to TALES TALES and/or TALES control de- fective	Check cable connec- tions. Check TALES control (reference voltage, FOC control, CV signal). Replace defective TALES.
	OK with dummy ter- mination	BTB/BC error	Use TestTools, BTB and BC tests.

Troubleshooting hints

Transmit power too low

- Check TX modules
- Check RFPA

Reflection too high

- Arcing of capacitors in BTB and/or BC
- Defective RF cables at BC, BTB and/or loose, corroded connections

Arcing

- Sporadic arcing -> BTB, otherwise BC defective
- Jumps in the reflection factor r with increasing transmit voltage
- $r > 70\%$ -> arcing in the BC

Normal conditions: reflection (voltage) < 50% (with BC)
 reflection (power) < 10%(with 50 Ohm dummy)

Note:

$$\text{reflection (volt)} = \sqrt{50 \text{ ohm} \times \text{reflection (watt)}}$$

- Arcing is sporadic: Arcing of C's in the BTB
(Capacitors in the BTB will be switched in various configurations)
- Arcing is permanent: Arcing of C's in BC

Oscillation of RFPA

- The reflection measured by TALES in a transmit voltage interval is significantly higher than the reflection factor measured by DICO.

Reason:

TALES measures wide-band, DICO measures narrow-band

- » Oscillation of RFPA; the RFPA also transmits at other frequencies besides the MR frequency.

Multinuclei Tests

Suggested RF Tests for Spectroscopy Option



For the RFPA- and RFAS test the cable to BCCS/X7 must be connected to TALES/X6 (refer to (→ Fig. 47 Page 89)

Otherwise the service plug and/or the coil plug 1 will be destroyed.

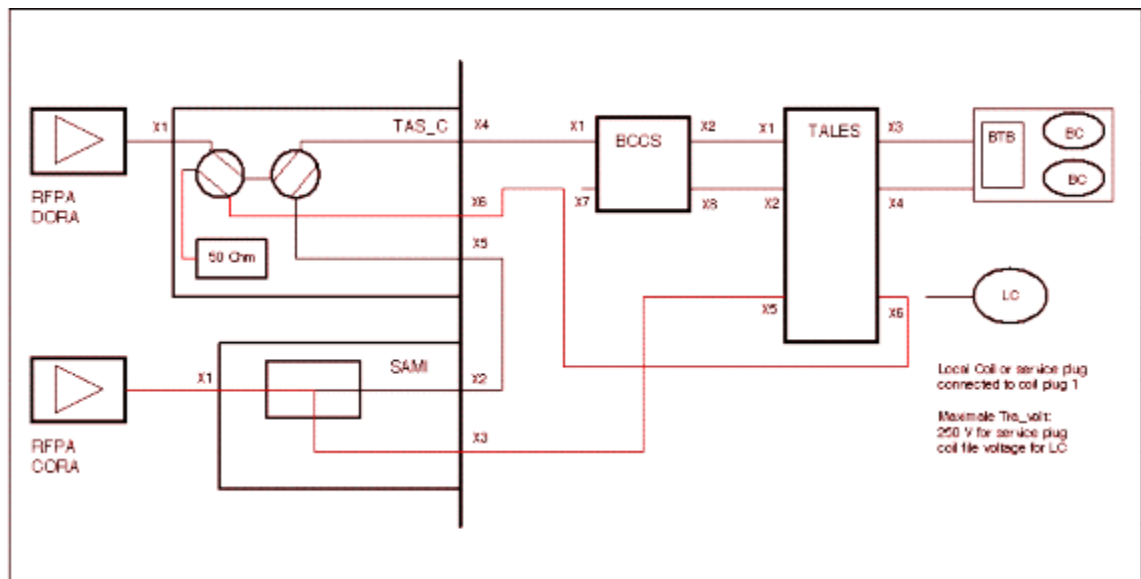
For the LCCS- and Cable tests the normal measuring configuration is used.

- Select the nucleus and checkboxes in the TestTools user interface as shown in the table below.

Test	Functional	Comm/Status	Interactive	Multinuclei	TX path
Mod/Rec 2			X	X	internal loop
RFPA		X		X	RFPA to dummy TAS_C
RFAS		X		X	RFPA to dummy TAS_C
LCCS ¹		X		X	TTX signal to service plug
Cable (LC TX path)				X	RFPA to service plug -> max.voltage 95V or to spectroscopy/x nucleus coil -> max. coil voltage

1. Abort the test after test completion of coil plug 1.

Fig. 47: LC TX path text configuration



MR frequencies (MHz):

DORA:

^1H 63.60

^{19}F 59.81

^3He 48.45

CORA:

^{31}P 25.74

^7Li 24.71

^{129}Xe 17.59

^{23}Na 16.82

^{13}C 15.99

Version 03:

- Chapter (Procedures), Section (RFPA) - troubleshooting procedure for tripped-off RFPA circuit breaker F11 added (→ Tripped-off Circuit Breaker F11 (DORA/ PORA) / Page 38)
- Chapter (Procedures), Section (RFPA) - troubleshooting procedure for CORA shut-downs added (→ CORA Shut-downs / Page 39) (CHARM_MR_00356099).

Version 04:

- All chapters - minor editorial changes.

Version 05:

- Chapter (Procedures) - Section (→ Check of 50 Ohm/Short Service Adapters (failed Tuning Calibration) / Page 66) CHARM TD_00015520).

There are no Hazard IDs in this document.

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